On the construction of Barnes-Wall lattices and their application in cryptography

Abstract

In this work, we investigate the application of Barnes-Wall lattices in post-quantum cryptographic schemes. We survey and analyze several constructions of Barnes-Wall lattices, including subgroup chains, the generalized k-ing construction, and connections with Reed-Muller codes, highlighting their equivalence over both $\mathbb{Z}[i]$ and $\mathbb{Z}$. Building on these structural insights, we introduce a new algorithm for efficient sampling from a discrete Gaussian distribution on BW_N lattices with $N = 2^n$. Our approach exploits the k-ing and squaring constructions to achieve low-variance sampling, which is particularly relevant for cryptographic applications such as digital signature schemes. We further examine the cryptographic hardness of Lattice Isomorphism Problem (LIP), showing that Barnes-Wall lattices provide inherent resistance to hull-based and other known attacks. Our results on sampling algorithms, combined with existing advances in the cryptanalysis of the LIP, indicate that Barnes-Wall lattices hold strong potential for the design of post-quantum schemes based on the LIP problem.

Keywords: Barnes–Wall lattice, Gaussian sampling, lattice-based cryptography, Lattice Isomorphism Problem, hull-attacks

1 Introduction

In recent years, the Lattice Isomorphism Problem (LIP) has attracted increasing attention in the field of post-quantum cryptography. It was first proposed as a foundation for constructing cryptographic schemes in works [1–4], where the authors demonstrated its potential. However, among all problems related to isometries, LIP remains the least studied. Current research in this area includes the analysis of possible attacks, such as methods based on graph isomorphisms and code equivalence [5, 6]. In particular, algorithms for solving the graph isomorphism problem, such as [7], and [8], can be adapted to analyze lattice structures. Additionally, in work [9], Babai proposed a quasipolynomial-time algorithm for graph isomorphism, which could also be applied to LIP.

Barnes–Wall lattices, first described in the context of sphere packing [10], represent an important class of lattices with unique properties, such as high density and unimodularity up to scaling [11]. These lattices can be constructed in various ways, including iterative constructions over the ring of Gaussian integers $\mathbb{Z}[i]$ and the use of Reed–Muller codes [12]. It is important to note that Barnes–Wall lattices have proven particularly useful in the context of LIP, as their structure allows for efficient implementation of sampling algorithms from Gaussian distributions [13]. Moreover, the use of these lattices makes hull-based attacks impossible, significantly enhancing their cryptographic security [14]. Thus, exploring the relationship between LIP and Barnes–Wall lattices opens new perspectives for developing robust cryptographic schemes.

Our contribution. In this work, we examine the lattice isomorphism problem in the context of Barnes–Wall lattices. We analyze various constructions of these lattices, including:

- Iterative construction via subgroup chains;
- k-ing construction, which generalizes the squaring construction;
- Construction based on Reed–Muller codes.

These constructions allow obtaining equivalent representations of Barnes–Wall lattices both over $\mathbb{Z}[i]$ and $\mathbb{Z}$, enabling the application of standard sampling algorithms developed for integer lattices.

The key result of this work is the development of an efficient sampling algorithm from the Gaussian distribution on BW_N lattices, tailored to their special structure. This algorithm leverages the idea of k-ing construction and enables sampling with a small standard deviation, which is crucial, for instance, for constructing secure and efficient digital signature schemes.

Organization of the paper. The structure of the article is organized to sequentially and logically reveal the key aspects of the study of the lattice isomorphism problem and its applications in the context of Barnes–Wall lattices. Section 2 provides basic definitions and notations, including a formal description of lattices, their properties, invariants, and the formulation of both the basic lattice isomorphism problem and its modifications. Section 3 examines several methods for constructing Barnes–Wall lattices and their specific features. Particular attention is paid to the k-ing construction, which generalizes the squaring construction and plays a key role in forming Barnes–Wall lattices. In Section 4 we present new results on optimizing the sampling process from distributions on lattices, along with their analysis for various parameters. The new sampling algorithm for Barnes–Wall lattices is proposed in Section 4.4. It utilizes the structure of Barnes–Wall lattices, enabling efficient sampling with complexity $\mathcal{O}(\dim(BW))$. Section 5 investigates the cryptographic security of the LIP on Barnes–Wall lattices. The concluding Section 6 discusses directions for further research, including potential applications of the proposed methods in quantum-resistant protocols.

2 Preliminaries

We will use the following notations:

- The sets of all integer, rational, real and complex numbers is denoted by $\mathbb{Z}, \mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ respectively.
- All vectors $\mathbf{x}$ and matrices $\mathbf{A}$ are denoted by bold lower and upper case letters respectively.
- The notation $x := y$ means that x is defined to be equal to y.
- We denote the set of unimodular matrices as $\mathcal{GL}_n(\mathbb{Z}) = \{\mathbf{U} \in \mathbb{Z}^{n\times n} : \det(\mathbf{U}) = \pm 1\}$.
- We denote the standard Euclidean inner product of two vectors $\mathbf{x} = (x_1, \ldots, x_k), \mathbf{y} = (y_1, \ldots, y_k) \in \mathbb{R}^k$ by $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^{k} x_i y_i$.
- The notation Λ_1/Λ_2 means the quotient of Λ_1 by Λ_2.
- The finite field consisting of q elements is denoted by $\mathbb{F}_q$.
- We denote the set of permutation matrices of size $n \times n$ with elements from $\mathbb{F}_q$ as $\mathrm{Perm}_n(\mathbb{F}_q)$.

2.1 Lattices

Informally, a *lattice* can be thought of as a regular, infinite grid of points in Euclidean space. More precisely, a lattice $\Lambda \subset \mathbb{R}^d$ is a discrete subgroup of the additive group $\mathbb{R}^d$. This means that:

- If $\mathbf{v}, \mathbf{w} \in \Lambda$ then also $\mathbf{v} \pm \mathbf{w} \in \Lambda$; that is, the set is closed under addition and subtraction.
- There exists a positive radius $r > 0$ such that for any two distinct lattice vectors $\mathbf{v}, \mathbf{w} \in \Lambda$, the Euclidean distance between them satisfies $\|\mathbf{v} - \mathbf{w}\| \geq r$. This condition ensures that the points of the lattice do not accumulate and thus form a discrete structure.

Here, $\|\cdot\|$ denotes the standard Euclidean *norm* on $\mathbb{R}^d$, defined by

$$\|\mathbf{x}\| = \|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^{d} x_i^2}.$$

A lattice can be described in terms of a finite set of linearly independent vectors, called a *basis*. Let $\mathbf{b}_0, \ldots, \mathbf{b}_{n-1} \in \mathbb{R}^d$ be linearly independent over $\mathbb{R}$. We arrange them as columns of the matrix $\mathbf{B} = [\mathbf{b}_0, \ldots, \mathbf{b}_{n-1}]$. The lattice generated by $\mathbf{B}$ is then defined as

$$\Lambda(\mathbf{B}) := \mathbf{B} \cdot \mathbb{Z}^n = \left\{ \sum_{i=0}^{n-1} x_i \mathbf{b}_i : x_i \in \mathbb{Z} \right\}.$$

The *rank* of a lattice Λ is the dimension of the real vector space spanned by its vectors:

$$\mathrm{rk}(\Lambda) := \dim(\mathrm{span}\,\Lambda).$$

Equivalently, the rank equals the number of basis vectors of Λ. A lattice $\Lambda \subset \mathbb{R}^d$ is called a full-rank lattice if its rank equals the dimension d of the vector space $\mathbb{R}^d$.

For a basis $\mathbf{B}$ of a lattice of rank n, all other bases have the form $\mathbf{B} \cdot \mathbf{U}$ for a unimodular matrix $\mathbf{U} \in \mathcal{GL}_n(\mathbb{Z})$.

Given a lattice Λ, one can define its *dual lattice*, which captures all vectors that have integer inner products with the lattice vectors. Formally,

$$\Lambda^\star = \{\mathbf{x} \in \operatorname{span}(\Lambda) : \langle \mathbf{x}, \mathbf{y} \rangle \in \mathbb{Z} \text{ for all } \mathbf{y} \in \Lambda\}.$$

For a lattice basis $\mathbf{B}$ of rank n, the *Gram matrix* is the symmetric, positive definite matrix consisting of all pairwise inner products of the basis vectors:

$$\mathbf{G} := \mathbf{B}^\top \mathbf{B} = (\langle \mathbf{b}_i, \mathbf{b}_j \rangle)_{0 \le i,j < n}.$$

The Gram matrix is a central object in analyzing the geometry of lattices, which is particularly important in cryptographic applications.

The *volume* (also called the determinant) of a lattice Λ is defined as

$$\operatorname{vol}(\Lambda) := \operatorname{vol}((\operatorname{span}\Lambda)/\Lambda) = (\det(\mathbf{B}^\top \mathbf{B}))^{1/2},$$

where $\mathbf{B}$ is a basis of Λ.

The volume measures the density of the lattice points in space. It is invariant under basis change and appears in hardness bounds (e.g., Minkowski's theorem) that connect geometry to cryptographic security parameters.

The *first minimum* of a lattice Λ is defined as the minimal norm of a non-zero lattice vector:

$$\lambda_1(\Lambda) := \min_{\mathbf{v} \in \Lambda \setminus \{\mathbf{0}\}} \{\|\mathbf{v}\|\}.$$

Many hard problems in lattice cryptography, such as SVP and GapSVP, are formulated in terms of approximating this quantity.

Theorem 1 (Minkowski's Theorem) *For a lattice Λ of rank n, the following bound holds for the first minimum:*

$$\lambda_1(\Lambda) \le \sqrt{n} \cdot \operatorname{vol}(\Lambda)^{1/n}.$$

Definition 1 (Hermite constant) The Hermite constant γ_n is defined as the maximum ratio

$$\gamma_n := \max_{\operatorname{rk}(\Lambda)=n} \left\{ \frac{\lambda_1(\Lambda)^2}{\operatorname{vol}(\Lambda)^{2/n}} \right\},$$

taken over all rank n lattices.

2.2 Lattice Isomorphism Problem

Definition 2 (Lattice Isomorphism Problem (LIP), Lattice Version) For given bases $\mathbf{B}, \mathbf{B}' \subset \mathbb{R}^n$ of two isomorphic lattices, compute an orthonormal transformation $\mathbf{O} \in O_n(\mathbb{R})$ and a unimodular transformation $\mathbf{U} \in \mathcal{GL}_n(\mathbb{Z})$ such that

$$\mathbf{B}' = \mathbf{O}\mathbf{B}\mathbf{U}.$$

A quadratic form of rank n is a positive definite symmetric real matrix $\mathbf{Q} \in S_n^{>0}(\mathbb{R})$. Then two quadratic forms $\mathbf{Q}, \mathbf{Q}' \in S_n^{>0}(\mathbb{R})$ are called *equivalent* ($\mathbf{Q} \simeq \mathbf{Q}'$) if there exists a unimodular transformation $\mathbf{U} \in \mathcal{GL}_n(\mathbb{Z})$ such that

$$\mathbf{Q}' = \mathbf{U}^\top \mathbf{Q}\mathbf{U}.$$

Denote the equivalence class of the quadratic form $\mathbf{Q}$ as

$$[\mathbf{Q}] := \{\mathbf{U}^\top \mathbf{Q}\mathbf{U} : \mathbf{U} \in \mathcal{GL}_n(\mathbb{Z})\}.$$

Definition 3 (Worst-Case Search Lattice Isomorphism Problem (wc-sLIP)) For a quadratic form $\mathbf{Q} \in S_n^{>0}$, the worst-case search lattice isomorphism problem (wc-sLIP) consists of finding a unimodular transformation $\mathbf{U} \in \mathcal{GL}_n(\mathbb{Z})$ such that

$$\mathbf{Q}' = \mathbf{U}^\top \mathbf{Q}\mathbf{U},$$

where $\mathbf{Q}' \in [\mathbf{Q}]$.

Note that this problem is equivalent to the original LIP problem, since we can extract an orthonormal transformation by computing $\mathbf{O} = \mathbf{B}'(\mathbf{B}\mathbf{U})^{-1}$.

Definition 4 (Worst-Case Distinguishing Lattice Isomorphism Problem wc-ΔLIP) For two quadratic forms $\mathbf{Q}_0, \mathbf{Q}_1 \in S_n^{>0}$, the worst-case distinguishing lattice isomorphism problem (wc-ΔLIP) involves the following: given an arbitrary quadratic form $\mathbf{Q}' \in [\mathbf{Q}_b]$, where $b \in \{0, 1\}$ is a uniformly random bit, it is required to recover b.

3 Barnes–Wall lattices and their constructions

In this section, we present various constructions of Barnes–Wall lattices. It is known that Barnes–Wall lattices can be obtained iteratively over the ring of Gaussian integers (see Section 3.1). In Sections 3.2 and 3.3, we present equivalent constructions over Gaussian integers and integers using so-called chains of partitions and Reed–Muller codes. These construction approaches enable us to define efficient Gaussian sampling for Barnes–Wall lattices in Section 4, including sampling over Gaussian integers.

3.1 Basic construction of Barnes–Wall lattices over Gaussian integers

A lattice defined as a free $\mathbb{Z}[i]$-module, where $\mathbb{Z}[i]$ is the ring of Gaussian integers, will be called a *complex lattice* and denoted by $\Lambda^{\mathbb{Z}[i]}$, and its generating matrix will be denoted by $\mathbf{B}^{\mathbb{Z}[i]}$. Let $\mathrm{rk}(\Lambda) = n$; then the generating matrix $\mathbf{B}$ of the corresponding integer lattice in $\mathbb{Z}^{2n}$ is obtained by associating each element $a + ib \in \mathbb{Z}[i]$ with the matrix

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix}.$$

The lattice generated by $\theta \cdot \mathbf{B}^{\mathbb{Z}[i]}$ for some $\theta \in \mathbb{Z}[i]$ will be denoted by $\theta\Lambda^{\mathbb{Z}[i]}$. Denote the integer part of the lattice $\theta\Lambda^{\mathbb{Z}[i]}$ by $\theta\Lambda$, and it is obtained from $\mathbf{B}$ as follows. Let $R(2,\theta)$ be the 2×2 matrix given by

$$R(2,\theta) = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \quad \text{where } \theta = 1+i.$$

The rotation operator $R(n,\theta)$ in dimension n is defined by applying $R(2,\theta)$ component-wise to the corresponding vector, i.e., it has the form $R(n,\theta) = \mathbf{I}_n \otimes R(2,\theta)$, where $\mathbf{I}_n$ is the $n \times n$ identity matrix, and $\otimes$ denotes the Kronecker product. Then the integer part $\theta\Lambda^{\mathbb{Z}[i]}$ is generated by the matrix $\mathbf{B} \cdot R(n,\theta)$.

Definition 5 (Barnes–Wall lattice) For any positive integer n, the Barnes–Wall lattice BW_N of dimension $N = 2^n$ over $\mathbb{Z}[i]$ is the lattice generated by the rows of the following matrix

$$BW_N = \begin{pmatrix} 1 & 1 \\ 0 & \theta \end{pmatrix}^{\otimes n},$$

or, using the recursive definition, we obtain an $N \times N$ matrix

$$BW_N = \begin{pmatrix} BW_{N/2} & BW_{N/2} \\ \mathbf{0} & \theta \cdot BW_{N/2} \end{pmatrix},$$

where $BW_1 = (1)$.

An important geometric property of Barnes–Wall lattices is that the minimal vector length in both the primal and dual lattices differs from the Minkowski bound by a factor of $\Theta(\sqrt[4]{n})$ [2]. In Section 3.3 we will show that BW_N can be alternatively defined as an integral lattice without the transition described above.

3.2 *k*-ing Barnes–Wall lattices construction

In [12], an approach was proposed to generate BW_N lattices with $N = 2^n$ using the squaring construction. Since the goal of our work is to develop a sampling algorithm for BW_N lattice, which, like any other lattice, is a discrete additive group, we will provide the description of the squaring construction directly in the context of groups.

Let S be a group and T its normal subgroup. Then partition of S via T is naturally induced by the quotient of S by T and is denoted as S/T. As a result, S can be expressed as the union of the disjoint cosets of this quotient, i.e. $S = \bigcup_{c \in [S/T]} c + T$, where $[S/T]$ is the set of representatives of S/T. Using this kind of partition, we can introduce the concept of a squaring construction of groups.

Definition 6 Let S be a group and T is its normal subgroup. Then the squaring construction $U = |S/T|^2$ is defined as the set of pairs (s_1, s_2) that belong to the same coset $c+T$. In other words, these are elements of the form

$$(s_1, s_2) = (c + t_1, c + t_2), \quad c \in [S/T], \quad t_i \in T,$$

where $[S/T]$ is the set of representatives of the cosets in the quotient group S/T.

In [12], a generalization of the squaring construction was also proposed, which we utilize in our sampling algorithm. Consider a chain of groups $G_n \subset \cdots \subset G_1 \subset G_0$. Then, the n-level squaring construction is defined as follows:

$$|G_0/G_1/\ldots/G_n|^N = ||G_0/G_1/\ldots/G_{n-1}|^{N/2}/|G_1/G_2/\ldots/G_n|^{N/2}|^2,$$

where $N = 2^n$ and $|\cdot|^N$ – n-level squaring construction notation.

The result obtained through this construction can be expressed as follows:

$$G_N = G_n^N + \sum_{0\le r\le n-1} G_{\mathcal{RM}}(r,n) \otimes [G_r/G_{r+1}],$$

where $G_{\mathcal{RM}}(r,n)$ denotes the set of all rows in $G_{(N,N)}$ with Hamming weight of at least 2^{n-r} and $G_{(N,N)} = G_{(2,2)}^{\otimes n}$ with $G_{(2,2)} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$.

Note that these constructions can be used to build Barnes–Wall lattices. In particular, the work [13] introduced a k-ing construction for lattice generation. In the case of Barnes–Wall lattices, it reduces to either the squaring construction ($k = 2$), or the two-level squaring construction ($k = 4$), which is a special case of the iterative squaring construction. Before discussing how they can be applied to Barnes–Wall lattices, we first provide a general description of the k-ing construction.

Definition 7 Let $L \subset M \subset S$ be a chain of full-rank lattices, and let $A = [S/M]$ and $B = [M/L]$ be sets of coset representatives for S/M and M/L respectively. The k-ing construction $\Gamma(L, A, B, k)$ is defined as follows

$$\Gamma(L, A, B, k) = \bigcup_{m\in A} \{\Gamma(L, B, k)_P + (m, \ldots, m)\},$$

where $\Gamma(L, B, k)_P = \{(t_1, \ldots, t_k) \,|\, t_i \in L + B, \sum_i t_i \in L\}$ is called parity-check sublattice.

Let us describe in more detail how to use the k-ing construction for building Barnes–Wall lattices BW_N with $N = 2^n$ in the case of $k = 4$. Consider the lattice chain

$$\theta^2 BW_{N/4} \subset \theta BW_{N/4} \subset BW_{N/4}$$

for $n \ge 2$. Let

$$A = [BW_{N/4}/\theta BW_{N/4}] \quad \text{and} \quad B = [\theta BW_{N/4}/\theta^2 BW_{N/4}]$$

be sets of coset representatives for the quotients $BW_{N/4}/\theta BW_{N/4}$ and $\theta BW_{N/4}/\theta^2 BW_{N/4}$ respectively. Then, BW_N can be constructed directly as the 4-ing construction $\Gamma(\theta^2 BW_{N/4}, A, B, 4)$ which is defined as follows:

$$BW_N = \Bigg\{(v_1 + t_1 + c, v_2 + t_2 + c, v_3 + t_3 + c, v_4 + t_4 + c),$$
$$v_k \in \theta^2 BW_{N/4}, c \in [BW_{N/4}/\theta BW_{N/4}], t_i \in [\theta BW_{N/4}/\theta^2 BW_{N/4}], \sum_{i=1}^{4} t_i = 0\Bigg\}.$$

In the context of [12], the Barnes–Wall lattices BW_N obtained through the 4-ing construction can also be expressed as a two-level squaring construction:

$$BW_N = |\theta^2 BW_{N/4}/\theta BW_{N/4}/BW_{N/4}|^4.$$

As we mentioned earlier, the k-ing construction with $k = 2$ can also be used to construct BW_N. In this case, it suffices to consider the lattice chain $\theta BW_{N/2} \subset BW_{N/2}$. Let $B = [BW_{N/2}/\theta BW_{N/2}]$ be the set of coset representatives for the quotient $BW_{N/2}/\theta BW_{N/2}$. Then BW_N is the parity-check lattice $\Gamma(\theta BW_{N/2}, B, 2)_P$ of the 2-ing construction, defined as follows:

$$BW_N = \Big\{(v_1 + m, v_2 + m), v_k \in \theta BW_{N/2}, m \in [BW_{N/2}/\theta BW_{N/2}]\Big\}.$$

Moreover, BW_N can also be expressed as a squaring construction $BW_N = |BW_{N/2}/\theta BW_{N/2}|^2$ as established in [12].

Thus, by applying the k-ing construction, BW_N can be iteratively constructed for any arbitrary n. Since the k-ing construction for $k = 2$ and $k = 4$ is closely related to the squaring construction, it is natural to generalize this approach for building BW_N. In [12], an iterative squaring construction was proposed for any $n > 0$ and $N = 2^n$ to build BW_N.

Let us consider a lattice chain $\theta^n BW_1 \subset \theta BW_1 \subset \cdots \subset BW_1$. Then, $BW_N = |BW_1/\theta BW_1/\ldots/\theta^n BW_1|^N$ represents a n-level iterative squaring construction. It enables the generation of the lattice BW_N as follows:

$$BW_N = \theta^n BW_1^N + \sum_{0 \le r \le n-1} G_{\mathcal{RM}}(r, n) \otimes [\theta^r BW_1/\theta^{r+1} BW_1].$$

Based on this iterative squaring construction of BW_N, later we will propose an iterative Gaussian sampling algorithm. However, to achieve this, we require the sets of coset representatives for the quotient $\theta^r BW_1/\theta^{r+1} BW_1$ for $0 \le r \le n-1$. Thus, we first determine the cardinality of these sets.

The following theorem [15] provides an answer to this question:

Theorem 2 *Let $\Lambda' \subseteq \Lambda$ be lattices. Then*
$$|\Lambda/\Lambda'| = \frac{\mathrm{vol}(\Lambda')}{\mathrm{vol}(\Lambda)}.$$

Now let us apply this theorem to Barnes–Wall lattices. We know that $\mathrm{vol}(BW_N) = \sqrt{N^N}$. Then, the volume of the scaled lattice θBW_N is $\mathrm{vol}(\theta BW_N) = \|\theta\|^n \,\mathrm{vol}(BW_N) = N \cdot \sqrt{N^N}$. As a result, we have

$$|BW_N/\theta BW_N| = N.$$

3.3 Barnes–Wall lattices over integers

Now we will give some equivalent constructions of the Barnes–Wall lattice over $\mathbb{Z}^m$. These constructions will be necessary later to justify the possibility of examining the security of schemes based on Gaussian Barnes–Wall lattices by transitioning to their integer versions with twice the original dimension.

Definition 8 (Reed–Muller Code) The Reed–Muller code $\mathcal{RM}(r,m)$ of order r and length 2^m is a binary code defined as:

$$\mathcal{RM}(r,m) = \begin{cases} \{0\}, & m=0, r<0, \\ \mathbb{F}_2, & m=0, r>0, \\ \mathbb{F}_2^{2^r}, & m=r, \\ \{(\mathbf{u}, \mathbf{u}+\mathbf{v}) : \mathbf{u} \in \mathcal{RM}(r, m-1), \mathbf{v} \in \mathcal{RM}(r-1, m-1)\}, & m>r. \end{cases}$$

The construction $(\mathbf{u}, \mathbf{u}+\mathbf{v})$, where $\mathbf{u} \in \mathcal{C}, \mathbf{v} \in \mathcal{C}'$, is called the Plotkin construction. The dimension of the Reed–Muller code $\mathcal{RM}(r,m)$ is $k = \sum_{j \le r} \binom{m}{j}$.

Let $\phi(\mathcal{C})$ denote the embedding of a linear code into $\mathbb{Z}^n$, mapping codewords over $\mathbb{F}_q$ to their corresponding vectors in the ring $\mathbb{Z}^n$.

Definition 9 (D-Construction) Let $\mathcal{C}_1 \subseteq \mathcal{C}_2 \subseteq \ldots \subseteq \mathcal{C}_L$, where each code $\mathcal{C}_j \subseteq \mathbb{F}_2^n$ is closed under the Schur product, i.e., $\mathbf{c}_j \star \mathbf{c}'_j = \left(c_{j,1}c'_{j,1}, \ldots, c_{j,n}c'_{j,n}\right) \in \mathcal{C}_{j+1}$ for all $\mathbf{c}_j, \mathbf{c}'_j \in \mathcal{C}_j$. Then D-construction lattices can be defined as follows

$$\Lambda_D = 2^L \mathbb{Z}^n + 2^{L-1}\phi\left(\mathcal{C}_L\right) + \ldots + 2\phi\left(\mathcal{C}_2\right) + \phi\left(\mathcal{C}_1\right).$$

In [12], it was proven that Barnes–Wall lattices can be obtained using the squaring construction

$$\Lambda\left(0,n\right) = \left|\Lambda\left(0,n-1\right)/R\Lambda\left(0,n-1\right)\right|^2,$$

where R is the rotation operator and

$$\Lambda(r,n) = R^{n-r}\mathbb{Z}^{2^{n+1}} + \sum_{r \le r' < n} G_{\mathcal{RM}}(r',n)\left[R^{r'-r}\mathbb{Z}^2/R^{r'-r+1}\mathbb{Z}^2\right],$$

$$BW_{N'}^{\mathbb{Z}} = \Lambda(0,n) = R^n\mathbb{Z}^{2^{n+1}} + \sum_{0 \le r' < n} G_{\mathcal{RM}}(r',n) \otimes \left[R^{r'}BW_2^{\mathbb{Z}}/R^{r'+1}BW_2^{\mathbb{Z}}\right],$$

where $BW^{\mathbb{Z}}_{N'} \cong BW_N$ (over Gaussian integers) with $N' = 2N$ and $N = 2^n$. This construction corresponds to the following chain of partitions:

$$BW^{\mathbb{Z}}_2/RBW^{\mathbb{Z}}_2/R^2BW^{\mathbb{Z}}_2/\ldots/R^nBW^{\mathbb{Z}}_2.$$

By transitioning to the 1-dimensional partition chain, we obtain

$$\Lambda(r,n) = \begin{cases} 2^{(n-r)/2}\mathbb{Z}^{2^{n+1}} + \sum_{\substack{r+1\le r'\le n \\ n-r' \text{ even}}} \mathcal{RM}(r',n+1)2^{(n-r')/2}, & \text{if } n-r \text{ is even,} \\ 2^{(n-r+1)/2}\mathbb{Z}^{2^{n+1}} + \sum_{\substack{r+1\le r'\le n \\ n-r' \text{ even}}} \mathcal{RM}(r',n+1)2^{(n-r')/2}, & \text{if } n-r \text{ is odd.} \end{cases}$$

This construction is clearly a D-construction and is equivalent to the well-known method of constructing integral Barnes–Wall lattices by applying this construction to Reed–Muller codes[16]:

$$BW^{\mathbb{Z}}_{N'} = \begin{cases} 2^{n/2}\mathbb{Z}^{2^{n+1}} + \sum_{\substack{1\le r\le n \\ n-r \text{ odd}}} \mathcal{RM}(r,n+1)2^{\frac{r-1}{2}}, & \text{if } n \text{ is even,} \\ 2^{(n+1)/2}\mathbb{Z}^{2^{n+1}} + \sum_{\substack{1\le r\le n \\ n-r \text{ even}}} \mathcal{RM}(r,n+1)2^{\frac{r-1}{2}}, & \text{if } n \text{ is odd.} \end{cases}$$

It is straightforward to show that this construction is a D-construction lattice, and thus, the sampling method from [16] can also be applied to Barnes–Wall lattices. For example, $BW^{\mathbb{Z}}_{128} = 8\mathbb{Z}^{128} + 4\mathcal{RM}(5,7) + 2\mathcal{RM}(3,7) + \mathcal{RM}(1,7)$.

The squaring and D-constructions yield Barnes–Wall lattices that are isomorphic to the original construction over Gaussian integers due to the isomorphism $\mathbb{Z}[i] \cong \mathbb{Z}^2 \cong BW_1 \cong BW^{\mathbb{Z}}_2$. Moreover, the squaring construction can be viewed from a coding theory perspective as the Plotkin $(\mathbf{u}|\mathbf{u}+\mathbf{v})$ construction. Let $\mathbf{u}'_2 = \mathbf{u}'_1 + \mathbf{v}$, $\mathbf{v} \in RBW^{\mathbb{Z}}_{N'/2}$ then:

$$\begin{aligned} BW^{\mathbb{Z}}_{N'} &= \{(\mathbf{u}'_1 + \mathbf{z}, \mathbf{u}'_2 + \mathbf{z}) \mid \mathbf{u}'_i \in RBW^{\mathbb{Z}}_{N'/2}, \mathbf{z} \in [BW^{\mathbb{Z}}_{N'/2}/RBW^{\mathbb{Z}}_{N'/2}]\} \\ &= \{(\mathbf{u}'_1 + \mathbf{z}, \mathbf{u}'_1 + \mathbf{z} + \mathbf{v})\} = \{(\mathbf{u}|\mathbf{u}+\mathbf{v}) \mid \mathbf{u} \in BW^{\mathbb{Z}}_{N'/2}, \mathbf{v} \in RBW^{\mathbb{Z}}_{N'/2}\}, \end{aligned}$$

where $\mathbf{R} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ is the rotation operator.

Using the latter approach to constructing Barnes–Wall lattices, the definition of the lattice's fundamental parameters becomes straightforward:

- Volume: $\mathrm{vol}(BW^{\mathbb{Z}}_{N'}) = \mathrm{vol}(BW^{\mathbb{Z}}_{N'/2}) \cdot \mathrm{vol}(R \cdot BW^{\mathbb{Z}}_{N'/2}) = 2^{n\cdot 2^{n-1}} = N^{N/2}$.
- Minimum distance: $\lambda_1(BW^{\mathbb{Z}}_{N'}) = d(BW^{\mathbb{Z}}_{N'}) = \sqrt{2^n} = \sqrt{N}$.
- Coding gain: $\gamma(BW^{\mathbb{Z}}_{N'}) = \frac{d^2}{\mathrm{vol}(BW^{\mathbb{Z}}_{N'})^{2/N'}} = \sqrt{N}$.

Thus, Barnes–Wall lattices over Gaussian integers are isomorphic to Barnes–Wall lattices over integers with doubled dimension. This result is crucial for cryptanalysis. For example, in Section 5.2.3 we show that this isomorphism can be used to prove the inapplicability of hull attacks for Barnes–Wall lattices over Gaussian integers.

4 Sampling algorithms for Barnes–Wall lattices

Sampling algorithms are an integral part of any lattice-based digital signature. For schemes based on the LIP problem, it is necessary to use lattices that allow efficient Gaussian sampling with small standard deviation. In this section, we present possible sampling algorithms for Barnes–Wall lattices for both the standard Euclidean norm and the so-called Q-norm used in the framework [2]. We also present new results on possible modifications to the sampling algorithm from [13]. In Section 4.4, we will present two new efficient iterative algorithms for discrete Gaussian sampling on Barnes-Wall lattices, enabling the selection of points from BW_N with $N = 2^n$ for arbitrary n. The first one is based on a k-ing sampler with $k = 4$ and $k = 2$ [13]. The second algorithm relies on the n-fold iterative squaring construction for Barnes-Wall lattices BW_N, described in Section 3.2.

4.1 Gaussian Sampling for q-ary Lattices from [16]

The paper [16] presents an efficient sampling method for q-ary structured lattices based on properties of error-correcting codes in the Lee metric. The authors prove the equivalence between sampling for $\mathbb{F}_q$-linear codes in the Lee metric and Gaussian sampling for lattices of constructions A, B, and D. This sampling is also can be efficient for Barnes–Wall lattices of small dimensions.

Lattice	Complexity Speedup	Sampling Width s Espitau, et al. [13]	Sampler [16]
A_2	$\approx 18\times$	$\approx \eta_\epsilon(A_2)$	$= \eta_\epsilon(A_2)$
E_8	$\approx 22\times$	$\approx \eta_\epsilon(E_8)$	$= \eta_\epsilon(E_8)$
D_n	$\approx 2\times$	$\approx \eta_\epsilon(D_n)$	$= \eta_\epsilon(D_n)$
Λ_{24}	$\approx 2\times$	$\approx \eta_\epsilon(E_8)$	$= \eta_\epsilon(\Lambda_{24})$
BW_N	$\approx 22\times$	$> \eta_\epsilon(BW_{N/2})$	$= \eta_\epsilon(BW_N)$

The authors of [16] demonstrate the possibility of improving sampling speed for BW_N lattices when $N < 64$ but for larger dimensions, no speedup can be achieved. The sampling method is based on finding codewords with a specific (sufficiently small) Hamming weight w_H and Lee weight $w_{\text{Lee}}(\mathbf{c}) = \sum_{j=1}^{N} \min\{c_j, q - c_j\}$. In the case of the Lee metric, it is necessary to find random vectors with a given *weight profile*. The weight profile in the Lee metric is used to define an analog of the *weight enumerator* used in the Hamming metric.

Definition 10 (Hamming Weight Enumerator) Let $\mathcal{C}$ be a linear $(n, k)_2$-code. The weight enumerator of $\mathcal{C}$ is defined as:

$$W_{\mathcal{C}}(x, y) = \sum_{\mathbf{c}\in\mathcal{C}} x^{n-w_{\mathrm{H}}(\mathbf{c})} y^{w_{\mathrm{H}}(\mathbf{c})} = \sum_{w=0}^{n} A_w x^{n-w} y^{w},$$

where $A_w(\mathcal{C}) = \#\{\mathbf{c} \in \mathcal{C} : w_{\mathrm{H}}(\mathbf{c}) = w\}$ and $0 \le w \le n$.

Definition 11 (Weight Profile) Let $\mathcal{C}$ be a linear code over $\mathbb{Z}_q$. The Lee weight profile of a codeword $\mathbf{c} \in \mathcal{C}$ is defined as:

$$[n_0(\mathbf{c}), n_1(\mathbf{c}), \ldots, n_\ell(\mathbf{c})], \quad \ell = \lceil q/2 \rceil,$$

where $n_w(\mathbf{c}) = \#\{i : w_{\text{Lee}}(c_i) = w\}$.

It is known that the problem of weight distribution (GWCP) in coding theory is NP-complete [17], as is the problem of finding codewords of a given weight in the Lee metric [18]. For Barnes–Wall lattices, the authors demonstrate the possibility of efficiently solving these problems in practice for dimensions $4 \le N \le 32$, including by using the structure of Reed–Muller codes and properties of the Schur product. For $N > 64$, the dimension of the Reed–Muller codes involved in the lattice construction becomes too large for efficient identification of codewords with a specific weight profile, making this sampling method inefficient for practical usage.

4.2 Q-norm and Euclidean norm sampling equivalence

In the framework [2], the authors transition from formulating the LIP problem over lattices to its description using the quadratic forms of these lattices. The framework employs Gaussian sampling in the so-called Q-norm. By considering the LIP problem through quadratic forms, it becomes possible to avoid using sampling algorithms specific to particular lattices employed in the scheme. Instead of sampling vectors belonging to the lattice, the authors propose sampling corresponding coefficient vectors from a Gaussian distribution over $\mathbb{Z}^n$. The relation between the two approaches can be defined as follows. Consider the vectors $\mathbf{u} = \mathbf{B}\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{v} = \mathbf{B}\mathbf{y} \in \mathbb{R}^n$, for some basis $\mathbf{B} \in \mathbb{R}^{n \times n}$, a quadratic form $\mathbf{Q} = \mathbf{B}^T\mathbf{B}$, and $\mathbf{x}, \mathbf{y} \in \mathbb{Z}^n$. Let us define

$$\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{Q}} = \mathbf{x}^T \mathbf{B}^T \mathbf{B} \mathbf{y} = (\mathbf{B}\mathbf{x})^T (\mathbf{B}\mathbf{y}) = \mathbf{u}^T \mathbf{v} = \langle \mathbf{u}, \mathbf{v} \rangle \in \mathbb{R}^1,$$

and

$$\|\mathbf{x}\|_{\mathbf{Q}} = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle_{\mathbf{Q}}} = \|\mathbf{u}\|_2.$$

In other words, instead of working with lattice vectors, we manipulate the integer coefficients. This method enables efficient signature verification. However, this comes at the cost of significantly reduced key generation efficiency compared with traditional Euclidean sampling. We now present definitions related to Gaussian sampling with respect to quadratic forms from [2].

For any quadratic form $\mathbf{Q} \in \mathcal{S}_n^{>0}$ we define *the Gaussian function* on $\mathbb{R}^n$ with parameter $s > 0$ and center $\mathbf{c} \in \mathbb{R}^n$ by

$$\rho_{\mathbf{Q},s,\mathbf{c}}(\mathbf{x}) = \exp(-\pi \|\mathbf{x} - \mathbf{c}\|_{\mathbf{Q}}^2 / s^2), \quad \forall \mathbf{x} \in \mathbb{R}^n.$$

The discrete Gaussian distribution is obtained in Ducas work also restricting the continuous Gaussian distribution to a discrete lattice. In the quadratic form setting, the discrete lattice will always be $\mathbb{Z}^n$, but with the geometry induced by the quadratic form. For any quadratic form $\mathbf{Q} \in \mathcal{S}_n^{>0}$, we define *the discrete Gaussian distribution* $D_{\mathbf{Q},s,\mathbf{c}}$ with center $\mathbf{c} \in \mathbb{R}^n$ and parameter $s > 0$ by

$$\Pr_{X \sim D_{\mathbf{Q},s,\mathbf{c}}}[X = \mathbf{x}] = \begin{cases} \frac{\rho_{\mathbf{Q},s,\mathbf{c}}(\mathbf{x})}{\rho_{\mathbf{Q},s,\mathbf{c}}(\mathbb{Z}^n)} & \text{if } \mathbf{x} \in \mathbb{Z}^n, \\ 0 & \text{otherwise.} \end{cases}$$

Definition 12 (Smoothing Parameter in $\mathbf{Q}$-norm) For a quadratic form $\mathbf{Q} \in \mathcal{S}_n^{>0}$ and $\epsilon > 0$, we define the smoothing parameter $\eta_\epsilon(\mathbf{Q})$ as the minimal $s > 0$ such that

$$\rho_{\mathbf{Q}^{-1},1/s}(\mathbb{Z}^n) \leq 1 + \epsilon.$$

The smoothing parameter is a central quantity for Gaussians over lattices, for example, it permits to control the variations of $\rho_{\mathbf{Q},s,\mathbf{c}}(\mathbb{Z}^n)$ over all centers $\mathbf{c}$.

Lemma 3 ([19]) *For any quadratic form $\mathbf{Q} \in \mathcal{S}_n^{>0}$, $\epsilon > 0$, center $\mathbf{c} \in \mathbb{R}^n$, and parameter $s > \eta_\epsilon(\mathbf{Q})$ we have*

$$(1-\epsilon)\frac{s^n}{\sqrt{\det(\mathbf{Q})}} \leq \rho_{\mathbf{Q},s,\mathbf{c}}(\mathbb{Z}^n) \leq (1+\epsilon)\frac{s^n}{\sqrt{\det(\mathbf{Q})}}.$$

Note that the smoothing parameter $\eta_\epsilon(\mathbf{Q})$ is an invariant property of the equivalence class $[\mathbf{Q}]$, and so we might also denote $\eta_\epsilon([\mathbf{Q}])$ for an equivalence class. While computing or even approximating the exact smoothing parameter is hard, we can obtain sufficient bounds via the dual form.

Lemma 4 (Smoothing bound [19]) *For any quadratic form $\mathbf{Q} \in \mathcal{S}_n^{>0}$, we have*

$$\eta_{2^{-n}}(\mathbf{Q}) \leq \sqrt{n}/\lambda_1(\mathbf{Q}^{-1}) \quad \text{and} \quad \eta_\epsilon(\mathbf{Q}) \leq \|\mathbf{B}^*_{\mathbf{Q}}\| \cdot \sqrt{\ln(2n(1+1/\epsilon))/\pi}$$

for $\epsilon > 0$.

Above the smoothing parameter, the discrete Gaussian distribution is in some sense "well-behaved" and we have the following tailbound that one would expect from a Gaussian distribution.

Lemma 5 (Tailbound [19]) *For any quadratic form $\mathbf{Q} \in \mathcal{S}_n^{>0}$, $\epsilon \in (0,1)$, center $\mathbf{c} \in \mathbb{R}^n$, and parameter $s \geq \eta_\epsilon(\mathbf{Q})$, we have*

$$\Pr_{\mathbf{x} \sim D_{\mathbf{Q},s,\mathbf{c}}}\left[\|\mathbf{x} - \mathbf{c}\|_{\mathbf{Q}} > s\sqrt{n}\right] \leq \frac{1+\epsilon}{1-\epsilon} \cdot 2^{-n}.$$

A constant factor above the smoothing parameter, we can furthermore lower bound the min-entropy of the distribution.

Lemma 6 (Min-entropy [20]) *For any quadratic form* $\mathbf{Q} \in \mathcal{S}_n^{>0}$*, positive* $\epsilon > 0$*, center* $\mathbf{c} \in \mathbb{R}^n$*, parameter* $s \geq 2\eta_\epsilon(\mathbf{Q})$*, and for every* $\mathbf{x} \in \mathbb{Z}^n$*, we have*

$$\Pr_{X \sim D_{\mathbf{Q},s,\mathbf{c}}}[X = \mathbf{x}] \leq \frac{1+\epsilon}{1-\epsilon} \cdot 2^{-n}.$$

Even when considering Barnes–Wall lattices over Gaussian integers, by transitioning to the Q-norm, we can employ slightly modified sampling algorithms for integral lattices, utilizing the idea described at the beginning of the section. In Algorithm 1, we present the GPV sampling [21] adapted for the Q-norm.

Algorithm 1 GPV sampler in Q-norm

Require: Ordered basis $\mathbf{B} = (\mathbf{b}_1, \ldots, \mathbf{b}_n)$ and Gram-Schmidt orthogonalization $\tilde{\mathbf{B}} = (\tilde{\mathbf{b}}_1, \ldots, \tilde{\mathbf{b}}_n)$ of a n-dimensional lattice Λ, parameter $s > 0$, and center $\mathbf{c} \in \mathbb{R}^n$
Ensure: Sample from discrete Gaussian distribution $D_{\Lambda,s,\mathbf{c}}$

```
v_n ← 0
c_n ← c
for i ← n to 1 do
    c'_i ← ⟨c_i, b̃_i⟩_Q / ⟨b̃_i, b̃_i⟩_Q
    s'_i ← s/‖b̃_i‖_Q
    Sample z_i ~ D_{Z, s'_i, c'_i}
    c_{i-1} ← c_i − z_i b_i
    v_{i-1} ← v_i + z_i b_i
end for
return v_0
```

Remark 1 The Gram–Schmidt orthogonalization process is also modified for use with the Q-norm. Specifically, the vector projections in the orthogonalization procedure must also be computed according to the Q-norm:

$$\mathrm{proj}_{\mathbf{u}}(\mathbf{v}) = \frac{\langle \mathbf{v}, \mathbf{u} \rangle_{\mathbf{Q}}}{\langle \mathbf{u}, \mathbf{u} \rangle_{\mathbf{Q}}} \mathbf{u}.$$

Although sampling in the Q-norm is possible for any lattice (even noninteger ones), an obvious drawback of this approach is the increased complexity. Indeed, to compute the inner product, and consequently vector projections, one needs to compute the product of a vector with a matrix:

$$\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{Q}} = \mathbf{x}^T \mathbf{Q} \mathbf{y},$$

instead of computing the inner product of two vectors. This implies that the complexity of all lattice algorithms in this case increases by a factor of $\mathcal{O}(n)$ compared to the Euclidean norm. Thus, we find that in the Q-norm, the complexity of Gram-Schmidt orthogonalization becomes $\mathcal{O}(n^4)$, and the complexity of computing the Hermite normal form (HNF) also becomes $\tilde{\mathcal{O}}(n^4)$ instead of $\tilde{\mathcal{O}}(n^3)$.

Thus, using sampling with the Q-norm in the framework [2] leads to a significant increase in key generation time, since it requires computing the Hermite normal form of the basis formed by $n \geq 256$ vectors. The speed of both generation and verification is worse compared to sampling with the Euclidean norm; however, the difference will not be as substantial as in the key generation process, since only a single vector needs to be sampled and verified.

This drawback can be resolved by using the Euclidean norm for signature verification. Consider the definition of Gaussian sampling in the Q-norm:

$$D_{\mathbf{Q},s,\mathbf{Ut}} = \frac{\rho_{\mathbf{Q},s,\mathbf{Ut}(\mathbf{x})}}{\rho_{\mathbf{Q},s,\mathbf{Ut}}(\mathbb{Z}^n)}$$

where the Gaussian function decomposes as:

$$\begin{aligned}
\rho_{\mathbf{Q},s,\mathbf{c}}(\mathbf{x}) &= \exp\left(-\pi\|\mathbf{x}-\mathbf{c}\|^2_{\mathbf{Q}}/s^2\right)\\
&= \exp\left(-\pi(\mathbf{x}-\mathbf{c})^\top\mathbf{Q}(\mathbf{x}-\mathbf{c})/s^2\right)\\
&= \exp\left(-\pi(\mathbf{x}-\mathbf{c})^\top\mathbf{B}^\top\mathbf{B}(\mathbf{x}-\mathbf{c})/s^2\right)\\
&= \exp\left(-\pi(\mathbf{Bx}-\mathbf{Bc})^\top(\mathbf{Bx}-\mathbf{Bc})/s^2\right)\\
&= \rho_{BW,s,\mathbf{Bc}}(\mathbf{x}').
\end{aligned}$$

Therefore, we obtain the equivalence $D_{\mathbf{Q},s,\mathbf{c}} \sim D_{BW,s,\mathbf{Bc}}(\mathbf{x}')$, where $\mathbf{x}' \in BW$. This shows that in the framework [2], it is possible to use Gaussian sampling from Barnes–Wall lattices with respect to the standard Euclidean norm. This means that we can utilize the advantages of transitioning to quadratic forms to simplify the description of the LIP problem and security proofs, while simultaneously using sampling in the Euclidean norm to improve the scheme's efficiency.

4.3 Gaussian Distribution on Quadratic Form Equivalence Classes

Based on the formulation of the LIP problem with respect to lattice quadratic forms described in Sections 2 and 3.3, it is necessary to present sampling algorithms from the Gaussian distribution on equivalence classes of quadratic forms, as an analog of computing a basis for an isomorphic lattice. First, we consider the classical algorithm for computing an equivalent lattice basis, originally described in [22].

Lemma 7 ([22, Lemma 7.1]) *There exists a polynomial-time algorithm that, given a lattice basis* $\mathbf{B}$ *and linearly independent lattice vectors* $\mathbf{V} \subset \Lambda(\mathbf{B})$ *with* $\|\mathbf{v}_1\| \leq \|\mathbf{v}_2\| \leq \ldots \leq \|\mathbf{v}_n\|$*, returns an equivalent basis* $\mathbf{R}$ *such that*

$$\|\mathbf{r}_k\| \leq \max\left\{(\sqrt{k}/2)\,\|\mathbf{v}_k\|, \|\mathbf{v}_k\|\right\}, \quad \forall k = 1,\ldots,n.$$

Moreover, the new basis satisfies

$$\operatorname{span}(\mathbf{r}_1, \dots, \mathbf{r}_k) = \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_k), \quad \|\mathbf{r}_k^*\| \le \|\mathbf{v}_k^*\| \text{ for all } k.$$

Since the vectors $\mathbf{V}$ belong to the lattice, we have $\mathbf{V} = \mathbf{BY}$, where $\mathbf{Y}$ is an integer square matrix. The matrix $\mathbf{Y}$ is non-singular but not necessarily unimodular, i.e., $\det(\mathbf{Y}) \in \mathbb{Z} \setminus \{0\}$. We transform $\mathbf{Y}$ into an upper triangular integer matrix by performing elementary row operations, while applying corresponding column operations to $\mathbf{B}$. This step is equivalent to finding a unimodular matrix $\mathbf{U}$ such that $\mathbf{H} = \mathbf{UY}$ is upper triangular. Let $\mathbf{R} = \mathbf{BU}^{-1}$. Since $\mathbf{U}$ is unimodular, $\mathbf{R}$ is a basis for $\Lambda(\mathbf{B})$. Furthermore, $\mathbf{V} = \mathbf{BU}^{-1}\mathbf{UY} = \mathbf{RH}$, hence $\operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_k) = \operatorname{span}(\mathbf{r}_1, \dots, \mathbf{r}_k)$ because $\mathbf{H}$ is upper triangular. Note that $\mathbf{v}_i^* = \mathbf{r}_i^* \mathbf{h}_{i,i}$, thus $\|\mathbf{r}_i^*\| = \|\mathbf{v}_i^*\|/|\mathbf{h}_{i,i}| \le \|\mathbf{v}_i^*\|$, since $\mathbf{h}_{i,i}$ is a non-zero integer.

Algorithm 2 Equivalent Basis Computation (Adapted from [22])

Require: Lattice basis $\mathbf{B}$ of $\Lambda(\mathbf{B})$, set of linearly independent lattice vectors $\mathbf{V} \subset \Lambda(\mathbf{B})$
Ensure: Basis $\mathbf{R}$ of Λ satisfying Lemma 7
1: Compute integer coefficients of vectors $\mathbf{V}$: $\mathbf{Y} = \mathbf{B}^{-1}\mathbf{V}$
2: Diagonalize $\mathbf{Y}$ using elementary operations, applying corresponding operations to $\mathbf{B}$, resulting in:
  Upper triangular matrix $\mathbf{H} = \mathbf{UY}$, $\mathbf{U} \in \mathcal{GL}_n(\mathbb{Z})$
  Equivalent basis $\mathbf{R} = \mathbf{BU}^{-1}$
3: **for** each vector $\mathbf{r}_i \in \mathbf{R}$ with $\|\mathbf{r}_i\| > \max\{\sqrt{i}/2, 1\} \cdot \|\mathbf{v}_i\|$ **do**
4:   Compute $\mathbf{R}^*$ – Gram-Schmidt orthogonalization of $\mathbf{R}$
5:   **if** $\mathbf{r}_i^* = \pm\mathbf{v}_i^*$ **then**
6:     Replace $\mathbf{r}_i$ with $\mathbf{v}_i$ in $\mathbf{R}$
7:   **else**
8:     Compute $\mathbf{v}' = \text{NearestPlane}(\mathbf{r}_i - \mathbf{r}_i^*, [\mathbf{r}_1, \dots, \mathbf{r}_{i-1}])$ ▷ [23]
9:     Replace $\mathbf{r}_i$ with $\mathbf{r}_i - \mathbf{v}'$
10:   **end if**
11: **end for**
12: Compute $\mathbf{U} = \mathbf{R}^{-1}\mathbf{B} \in \mathcal{GL}_n(\mathbb{Z})$
13: Return pair $(\mathbf{R}, \mathbf{U})$

The diagonalization process in Step 2 of Algorithm 2 can be performed using any algorithm for computing the Hermite Normal Form (HNF) of an integer matrix.

In [2], the authors adapt Algorithm 2 for the Q-norm and quadratic forms setting. However, as demonstrated in Section 3.3, this adaptation results in substantially slower key generation when applied to Barnes–Wall lattices in cryptographic schemes. Let $\mathbf{Q}$ be the Gram matrix of basis $\mathbf{B} \in \mathbb{R}^{n \times n}$ for a full-rank lattice Λ. Sampling from the Gaussian distribution $\mathcal{D}_s([\mathbf{Q}])$ can be performed using Algorithm 3.

The choice of parameter s depends on the specific Gaussian sampling algorithm. To minimize the number of repetitions of step 3 in Algorithm 3, we impose the constraint

Algorithm 3 Gaussian Sampling $\mathcal{D}_s([\mathbf{Q}])$ [2]

Require: Quadratic form $\mathbf{Q} \in \mathcal{S}_n^{>0}$ corresponding to lattice basis $\mathbf{B} \in \mathbb{R}^{n\times n}$ of Λ, parameter $s \geq \max\{\lambda_n(\mathbf{Q}), \eta_\epsilon(\mathbf{Q})\}$

Ensure: Quadratic form $\mathbf{P} = \mathbf{U}^T\mathbf{QU}$, matrix $\mathbf{U}$

Set $\hat{m} = \left\lceil \frac{2n}{1-(1+e^{-\pi})^{-1}} \right\rceil$

while rank$(\mathbf{Y}') \neq n$ **do**

  $\mathbf{Y}' \leftarrow (\mathbf{y}_1, \ldots, \mathbf{y}_{\hat{m}}) \in \mathcal{D}^{\hat{m}}_{\mathbf{Q},0,s}$

end while

Select first n linearly independent vectors $\mathbf{Y}$ from $\mathbf{Y}'$

$\mathbf{V} \leftarrow \mathbf{BY}$

Compute $(\mathbf{R}, \mathbf{U})$ using Algorithm 2

$\mathbf{P} \leftarrow \mathbf{R}^\top\mathbf{R}$ ▷ $\mathbf{P} = \mathbf{U}^\top\mathbf{QU}$

Return $(\mathbf{P}, \mathbf{U})$

$s \geq \lambda_n(\mathbf{Q})$. Thus, if rank$(\mathbf{y}_1, \ldots, \mathbf{y}_i) < n$, then since $s \geq \lambda_n(\mathbf{Q})$, it follows from [24] that the next sampled vector $\mathbf{y}_{i+1} \leftarrow D_{\mathbf{Q},s}$ does not belong to the span of $\mathbf{y}_1, \ldots, \mathbf{y}_i$ with probability at least $1 - (1 + e^{-\pi})^{-1} > 0$.

For $\hat{m} = \left\lceil \frac{2n}{1-(1+e^{-\pi})^{-1}} \right\rceil$, the expected number of iterations $E[T]$ is bounded by the expectation of a geometric distribution with success probability $1 - p_{\text{fail}}$, i.e.,

$$E[T] \leq \frac{1}{1 - p_{\text{fail}}} \leq \frac{1}{1 - e^{-2}} < 2.$$

Thus, Algorithm 3 runs in polynomial time when

$$s \geq \max\left\{\lambda_n(\mathbf{Q}), \eta_\epsilon(\mathbf{Q})\right\}.$$

The work [2] proves the correctness of this sampling procedure and shows the distribution's independence from the specific representative of the equivalence class $\mathbf{Q}' \in [\mathbf{Q}]$. The transformation $\mathbf{U}$ obtained by the algorithm is uniformly distributed on the set of isomorphisms between quadratic forms $\mathbf{Q}$ and $\mathbf{P}$.

4.4 Novel sampling algorithms for Barnes–Wall Lattices

***k*-ing sampler**

The work [13] introduced a k-ing sampling algorithm for lattices that can be constructed using the k-ing construction. As demonstrated earlier, this class of lattices includes BW_N. Thus, we will firstly describe the k-ing sampling algorithm for the cases $k = 4$ (Algorithm 4) and $k = 2$ (Algorithm 5). Then, we present an iterative k-ing sampling algorithm capable of sampling elements from BW_N with $N = 2^n$ for arbitrary $n \geq 1$ by leveraging samples from the scaled lattices $\theta^s BW_1$ for $0 \leq s \leq n$.

Consider the lattice chain

$$\theta^2 BW_{N/4} \subset \theta BW_{N/4} \subset BW_{N/4}.$$

Let $A = [BW_{N/4}/\theta BW_{N/4}]$ and $B = [\theta BW_{N/4}/\theta^2 BW_{N/4}]$ be the sets of coset representatives $BW_{N/4}/\theta BW_{N/4}$ and $\theta BW_{N/4}/\theta^2 BW_{N/4}$ respectively. By definition of the k-ing construction, we have

$$BW_N = \bigcup_{m \in A} \{\Gamma(\theta^2 BW_{N/4}, B, 4) + (m, m, m, m)\}.$$

Thus, lattice sampling based on this approach reduces to two main steps:

1. Uniformly random sampling of a coset representative m from $[BW_{N/4}/\theta BW_{N/4}]$;
2. Sampling in the sublattice $\Gamma(\theta^2 BW_{N/4}, B, 4)_P$.

Moreover, by utilizing the exact sequence

$$0 \longrightarrow \theta^2 BW_{N/4}^4 \longrightarrow \Gamma(\theta^2 BW_{N/4}, B, 4)_P \longrightarrow \Gamma(\theta^2 BW_{N/4}, B, 4)_P/\theta^2 BW_{N/4}^4 \longrightarrow 0$$

and the short exact sequence sampling algorithm introduced in [13], we can reduce the sampling problem for the parity-check sublattice $\Gamma(\theta BW_{N/4}, B, 4)_P$ to another two steps:

1. Uniformly random selection of $t_i \in B$ for $1 \leq i \leq 4$, such that $\sum_{i=1}^{4} t_i = 0$;
2. Sampling elements $u_i \in \theta^2 BW_{N/4}$ from a discrete Gaussian distribution.

Below we present the k-ing sampling Algorithm 4 adapted for Barnes–Wall lattices BW_N in the case $k = 4$.

Algorithm 4 k-ing sampling algorithm for BW_N in the case $k = 4$

Require: A chain of full-rank lattices $\theta^2 BW_{N/4} \subsetneq \theta BW_{N/4} \subsetneq BW_{N/4}$, sets of coset representatives $A = \left[BW_{N/4}/\theta BW_{N/4}\right]$ and $B = \left[\theta BW_{N/4}/\theta^2 BW_{N/4}\right]$, a sampler $\mathcal{D}_{\theta^2 BW_{N/4},\cdot,\Sigma}$ with covariance $\Sigma > \eta_\epsilon(\theta^2 BW_{N/4})$, center $\mathbf{c} \in BW_{N/4}^4 \otimes \mathbb{R}$.
Ensure: $\mathbf{v}$ is statistically close to $\mathcal{D}_{\Gamma\left(\theta^2 BW_{N/4}, A, B, 4\right), \mathbf{c}, \Sigma}$.

$\alpha := U(A)$. ▷ $U(A)$ means uniform sampling of elements from A
$\mathbf{m} := (\alpha, \alpha, \alpha, \alpha)$.
for $i = 1, \dots 3$ **do**
  $t_i := U(B)$
end for
$t_4 := -\sum_{i=1}^{3} t_i$.
$\mathbf{t} := (t_1, \dots, t_4)$.
$\mathbf{u} := \mathcal{D}_{\theta^2 BW_{N/4}^4, \mathbf{c}-\mathbf{t}-\mathbf{m}, \Sigma}$.
return $\mathbf{u} + \mathbf{t} + \mathbf{m}$.

Note that this sampling algorithm can also be adapted for case $k = 2$. To achieve this, it suffices to consider the lattice chain $\theta BW_{N/2} \subsetneq BW_{N/2}$ and the short exact sequence

$$0 \longrightarrow \theta BW_{N/2}^2 \longrightarrow \Gamma(\theta BW_{N/2}, B, 2)_P \longrightarrow \Gamma(\theta BW_{N/2}, B, 2)_P/\theta BW_{N/2}^2 \longrightarrow 0.$$

In other words, it suffices to remove all steps unrelated to sampling in the parity-check sublattice. Since for $k = 2$, the Barnes–Wall lattice satisfies $BW_N = \Gamma(\theta BW_{N/2}, B, 2)_P$, this yields the following Algorithm 5.

Algorithm 5 k-ing sampling algorithm for BW_N in the case $k = 2$

Require: A chain of full-rank lattices $\theta BW_{N/2} \subsetneq BW_{N/2}$, set of coset representatives $B = [BW_{N/2}/\theta BW_{N/2}]$ of quotient $BW_{N/2}/\theta BW_{N/2}$, sampler $\mathcal{D}_{\theta BW_{N/2},\cdot,\Sigma}$ with covariance $\Sigma > \eta_\epsilon(\theta BW_{N/2})$, center $\mathbf{c} \in BW_{N/2}^2 \otimes \mathbb{R}$.

Ensure: $\mathbf{v}$ is statistically close to $\mathcal{D}_{\Gamma\left(\theta BW_{N/2},B,2\right)_P,\mathbf{c},\Sigma}$.

1: $t_1 := U(B)$

2: $t_2 := -t_1$.

3: $\mathbf{t} := (t_1, t_2)$.

4: $\mathbf{u} := \mathcal{D}_{\theta BW_{N/2}^2,\mathbf{c}-\mathbf{t},\Sigma}$.

5: **return** $\mathbf{u} + \mathbf{t}$.

As we can see, these algorithms enable sampling elements from the Barnes–Wall lattice BW_N by using $BW_{N/2}$ and its sublattices in the case of $k = 4$, and $BW_{N/2}$ and its sublattices in the case of $k = 2$. This raises a natural question: can we sample elements from BW_N directly using the lattice BW_1? To address this, we present two novel discrete Gaussian sampling algorithms for Barnes–Wall lattices. The first algorithm is based on the iterative application of the k-ing sampler for $k = 4$ and $k = 2$. Another algorithm leverages the n-level iterative squaring construction.

Let us begin with the recursive k-ing sampling algorithm. It takes as input parameter $N = 2^n$, the dimension of the target lattice BW_N, and $M = 2^m$, the dimension of the lattice BW_M on the current iteration step ($M = N$ on the first iteration step). Additionally, the algorithm takes center $\mathbf{c} \subseteq \theta^n BW_1 \otimes \mathbb{R}$ and covariance $\Sigma > \eta_\epsilon(\theta^n BW_N)$.

The algorithm iteratively goes down from the initial lattice BW_N to lattice BW_1 using k-ing samplers for $k = 4$ and $k = 2$. For the sampling from lattice BW_1, we apply Klein's sampler [25]. The correctness of this algorithm is based on correctness of the correctness of k-ing sampler introduced in [13]. The complete iterative k-ing sampling algorithm is presented below Algorithm 6.

Sampler based on iterative squaring construction

The second sampling Algorithm 7 we propose in this work enables sampling elements from the BW_N with $N = 2^n$ for $n \geq 1$ using only the lattice BW_1 and its sublattices. It gives us a discrete Gaussian sampling algorithm based on the iterative quadratic construction. As demonstrated earlier, the Barnes–Wall lattice admits the

Algorithm 6 iterative_king_sampler: recursive k-ing sampling algorithm for BW_N

Require: $n \in \mathbb{Z}_{>0}$ such that $N = 2^n$ is the dimension of the target lattice BW_N, $k \in \mathbb{Z}$, where $k = 0$ on the first iteration step, covariance $\Sigma > \eta_\epsilon(\theta^n BW_1)$, center $\mathbf{c} \subseteq \theta^n BW_1^N \otimes \mathbb{R}$.

Ensure: $\mathbf{v} \in BW_N$ is statistically close to $\mathcal{D}_{BW_N, \mathbf{c}, \Sigma}$.

```
if n mod 2 = 1 then
    v_1 = U([BW_{N/2}/θBW_{N/2}])
    v = (v_1, −v_1)
    u = iterative_king_sampler(n, k + 1, Σ, c − v)          ▷ u ∈ θBW^2_{N/2}
    return u + v
else
    for j = 0 to 2^k do
        γ_j = U([θ^k BW_{N/2^k}/θ^{k+1} BW_{N/2^k}])
        m_j = (γ_j, γ_j, γ_j, γ_j)
        for s=1 to 3 do
            t_{j,s} = U([θ^{k+1} BW_{N/2^k}/θ^{k+2} BW_{N/2^k}])
            t_j = (t_{j,1}, t_{j,2}, t_{j,3}, −∑_{s=1}^{3} t_{j,s})
        end for
        if k+2==n then
            u_j = Klein_Sampler(θ^n BW_1^4, Σ, c − m − t)          ▷ [25]
        else
            u = iterative_king_sampler(n, k + 2, Σ, c − m − t)
        end if
    end for
    m = (m_1, m_2, . . . , m_{2^k})
    t = (t_1, t_2, . . . , t_{2^k})
    u = (u_1, u_2, . . . , u_{2^k})
    return u + m + t
end if
```

decomposition:

$$BW_N = \theta^n BW_1^N + \sum_{0 \le r \le n-1} G_{\mathcal{RM}}(r, n) \otimes [\theta^r BW_1/\theta^{r+1} BW_1].$$

This Barnes–Wall lattice construction enables the creation of a sampling algorithm analogous to the k-ing sampling approach. The algorithm takes as inputs lattice chain $\theta^n BW_1 \subset \theta BW_1 \subset \cdots \subset BW_1$, sets of coset representatives $A_j = [\theta^j BW_1/\theta^{j+1} BW_1]$ of quotients $\theta^j BW_1/\theta^{j+1} BW_1$ for $j = 0, \ldots, n-1$, a sampler $\mathcal{D}_{BW_1, \cdot, \Sigma}$ with covariance $\Sigma > \eta_\epsilon(BW_1)$ and center $\mathbf{c} \in BW_1^N \otimes \mathbb{R}$.

The iterative sampling Algorithm 7 consists of two main parts:

1. Uniform random selection of an element c_j from each coset representative set A_j;
2. Sampling of a vector $\mathbf{u} = (u_1, u_2, \ldots, u_N)$ where each $u_i \sim \mathcal{D}_{BW_1, \cdot, \Sigma}$.

Note, that Klein's sampling algorithm [25] can be used as the sampler $\mathcal{D}_{BW_1,\cdot,\Sigma}$. Let $N = 2^n$ be the dimension of the lattice BW_N. Then, Algorithm 7 has complexity $O(N)$. Thus, according to Table 4.4, this algorithm is the most efficient sampling algorithm for BW_N lattices.

Table 1 A comparison of sampling algorithms for lattice BW_N with $N = 2^n$.

Sampling algorithm	Complexity of algorithm	Accumulated statistic distance δ_n	Smoothing parameter
Klein sampler [25, 26]	$O(N^2)$	ϵ	$\eta_\Sigma(BW_N)$
n-level squaring sampler [7]	$O(N)$	2ϵ	$\eta_\Sigma((1+i)^n BW_1)$
Iterative king sampler [6]	$O\left(2^{\frac{1}{2}\log^2 N}\right)$	2ϵ	$\eta_\Sigma((1+i)^n BW_1)$

Algorithm 7 n-level squaring sampler

Require: A chain of lattices $\theta^n BW_1 \subset \theta^{n-1} BW_1 \cdots \subset \theta BW_1 \subset BW_1$, sets of coset representatives $A_j = [\theta^j BW_1/\theta^{j+1} BW_1]$ of $\theta^j BW_1/\theta^{j+1} BW_1$ for $j = 0, \ldots, n-1$, sampler $\mathcal{D}_{BW_1,\cdot,\Sigma}$ with covariance $\Sigma > \eta_\epsilon(\theta^n BW_1)$, center $\mathbf{c} \in BW_1^N \otimes \mathbb{R}$.

Ensure: $\mathbf{v} \in BW_N$ is statistically close to $\mathcal{D}_{BW_N,\mathbf{c},\Sigma}$.

for $j = n-1$ to 0 **do**

  $\alpha_j := U(A_j)$.

  $\mathbf{w}_j = G_{\mathcal{RM}}(j, n) \otimes \alpha_j$.

end for

$\mathbf{u} := \mathcal{D}_{(\theta^n BW_1)^N,\mathbf{c}-\sum_{j=1}^{n-1}\mathbf{w}_j,\Sigma}$.

return $\mathbf{u} + \sum_{j=0}^{n-1} \mathbf{w}_j$.

Theorem 8 (Correctness of the Algorithm 7) *When $\Sigma \succ \eta_\epsilon(\theta^n BW_1)$, Algorithm 7 is correct. Moreover, let $\mathfrak{D}$ be the distribution of its output. For $\epsilon < \frac{1}{2}$, we have*

$$\sup \left| \frac{\mathfrak{D}(\mathbf{v})}{\mathcal{D}_{BW_N,\mathbf{c},\Sigma}(\mathbf{v})} - 1 \right| \leq 4\epsilon.$$

In particular, $\mathfrak{D}$ is within statistical distance 2ϵ of $\mathcal{D}_{BW_N,\mathbf{c},\Sigma}$.

Proof Let $\epsilon > 0$ and $\mathbf{v} \in BW_N$ with $N = 2^n$. Consider a chain of lattices $\theta^n BW_1 \subset \theta^{n-1} BW_1 \subset \cdots \subset \theta BW_1 \subset BW_1$. The output of the algorithm shows that $\mathbf{v} = \mathbf{u} + \mathbf{w}_0 + \mathbf{w}_1 + \cdots + \mathbf{w}_{n-1}$, where $\mathbf{w}_i = G_{RM}(j, n) \otimes \alpha_j$ with $a_j \in A_j = [\theta^j BW_1/\theta^{j+1} BW_1]$ and $j = 0, \ldots, n-1$.

We note that α_j is chosen uniformly at random from the set of representatives of the quotient $\theta^j BW_1/\theta^{j+1}BW_1$. Since $\mathbf{w}_j = G_{RM}(j,n)\otimes\alpha_j$, the choice of $\mathbf{w}_j$ depends only on the choice of $\alpha_j \in A_j$. Therefore, by the law of total probability, we have:

$$\mathfrak{D}(\mathbf{v}) = \Pr\left(\mathbf{u} = \mathbf{v} - \sum_{j=0}^{n-1}\mathbf{w}_j \,\middle|\, \bigcap_{j=0}^{n-1}\{\alpha_j \in A_j, \mathbf{w}_j = G_{RM}(j,n)\otimes\alpha_j\}\right)\cdot$$

$$\cdot \Pr\left(\{\alpha_{n-1}\in A_{n-1}, \mathbf{w}_{n-1} = G_{RM}(n-1,n)\otimes\alpha_{n-1}\} \,\middle|\, \bigcap_{k=0}^{n-2}\{\alpha_j \in A_j, \mathbf{w}_j = G_{RM}(j,n)\otimes\alpha_j\}\right)\cdot\ldots$$

$$\ldots\cdot\Pr(\{\alpha_0\in A_0, \mathbf{w}_0 = G_{RM}(0,n)\otimes\alpha_0\})$$

Since all α_k are choosen independently, for $k = n-1,\ldots,1$, we have:

$$\Pr\left(\{\alpha_k\in A_k, \mathbf{w}_k = G_{RM}(k,n)\otimes\alpha_k\} \,\middle|\, \bigcap_{l=0}^{k-1}\{\alpha_l \in A_l, \mathbf{w}_l = G_{RM}(l,n)\otimes\alpha_l\}\right) =$$
$$= \Pr\left(\{\alpha_k\in A_k, \mathbf{w}_k = G_{RM}(k,n)\otimes\alpha_k\}\right)$$

In algorithm 7 α_j are chosen uniformly at random from the set of representatives A_j. Hence, we have:

$$\Pr\left(\{\alpha_k\in A_k, \mathbf{w}_k = G_{RM}(k,n)\otimes\alpha_k\}\right) = \frac{1}{|A_k|} = \frac{1}{2}$$

Now consider

$$\Pr\left(\mathbf{u} = \mathbf{v} - \sum_{j=1}^{n}\mathbf{w}_j \,\middle|\, \bigcap_{k=0}^{n-1}\{\alpha_j \in A_j, \mathbf{w}_j = G_{RM}(j,n)\otimes\alpha_j\}\right)$$

Let $\pi_j : \theta^j BW_1 \longrightarrow \theta^j BW_1/\theta^{j+1}BW_1$. It is easy to see that $\mathbf{w} = \sum_{j=1}^{n}\mathbf{w}_j \in BW_N$. Moreover, we can construct an isomorphism

$$\phi : BW_N/(1+i)^n BW_1^N \longrightarrow BW_1^N/(1+i)BW_1^N \times\cdots\times (1+i)^{n-1}BW_1^N/(1+i)^n BW_1^N$$
$$\phi(\mathbf{w} \mod (1+i)^n BW_1^N) \mapsto (\mathbf{w}_1 \mod (1+i)BW_1^N, \ldots, \mathbf{w}_n \mod (1+i)^n BW_1^N)$$

This means that $\mathbf{w}$ is a representative of a coset $q \in BW_N/(1+i)^n BW_1^N$ such that $\phi(q) \mapsto (\pi_1'(\mathbf{w}_0), \pi_1'(\mathbf{w}_1), \ldots, \pi_{n-1}'(\mathbf{w}_{n-1}))$, where $\pi_j'(\mathbf{w}_j) = \pi_j(\alpha_j)\otimes G_{RM}(j,n)$ for $j = 0,\ldots,n-1$. As a result, choosing such a vector $\mathbf{w} \in [BW_N/(1+i)^n BW_1^N]$, where $\mathbf{w} = \sum_{j=0}^{n-1}\mathbf{w}_j$ is equivalent to chose each $\mathbf{w}_j \in A_j$. Therefore, we conclude that

$$\Pr\left(\mathbf{u} = \mathbf{v} - \sum_{j=0}^{n-1}\mathbf{w}_j \,\middle|\, \bigcap_{j=0}^{n-1}\{\alpha_j \in A_j, \mathbf{w}_j = G_{RM}(j,n)\otimes\alpha_j\}\right) = Pr(\mathbf{u} = \mathbf{v} - \mathbf{w} | \pi(\mathbf{w}) = q),$$

where $\pi : BW_N \longrightarrow BW_N/(1+i)^n BW_1^N$.

$$Pr(\mathbf{u} = \mathbf{v} - \mathbf{w}|\pi(\mathbf{w}) = q) = \frac{\rho_\Sigma(\mathbf{v} - \mathbf{w} - (Id-\pi)(\mathbf{c}-\mathbf{w}))}{\rho_\Sigma((1+i)^n BW_1^N - (Id-\pi)(\mathbf{c}-\mathbf{w}))} =$$
$$= \frac{\rho_\Sigma((Id-\pi)(\mathbf{v}-\mathbf{c}))}{\rho_\Sigma((1+i)^n BW_1^N - (Id-\pi)(\mathbf{c}-\mathbf{w}))}$$

From [13], we know that for $\Sigma \succ \eta_\epsilon(\theta^n BW_1^N) \geq \eta_\epsilon(\overline{\theta^n BW_1^N})$, the denominator satisfies

$$\rho_\Sigma(\theta^n BW_1^N - (Id-\pi)(\mathbf{c}-\mathbf{w})) \in$$

$$\in \left[1, \frac{1+\epsilon}{1-\epsilon}\right] \frac{1}{|BW_N/(1+i)^n BW_1^N|} \rho_\Sigma(\overline{\theta^n BW_N} - (Id-\pi)(\mathbf{c}-\mathbf{w})) \in$$
$$\in \left[\frac{1-\epsilon}{1+\epsilon}, \left(\frac{1+\epsilon}{1-\epsilon}\right)^2\right] \frac{1}{|BW_N/(1+i)^n BW_1^N|} \rho_\Sigma(\overline{\theta^n BW_N} - (Id-\pi)(\mathbf{c})).$$

Here, $\overline{\theta^n BW_N} = span(\theta^n BW_N, |BW_N/(1+i)^n BW_1^N|) = BW_N$. Moreover, $|BW_N/(1+i)^n BW_1^N| = \prod_{j=0}^{n-1} A_j = 2^n$. Finally we obtain:

$$\mathfrak{D}(\mathbf{v}) = \left[\left(\frac{1-\epsilon}{1+\epsilon}\right)^2, \left(\frac{1+\epsilon}{1-\epsilon}\right)\right] \cdot \frac{\rho_\Sigma((Id-\pi)(\mathbf{v}-\mathbf{c}))}{\rho_\Sigma(BW_N - (Id-\pi)(\mathbf{c}))} =$$
$$= \left[\left(\frac{1-\epsilon}{1+\epsilon}\right)^2, \left(\frac{1+\epsilon}{1-\epsilon}\right)\right] \cdot \mathcal{D}_{BW_N,\mathbf{c},\Sigma}(\mathbf{v})$$

Therefore

$$\left(\frac{1-\epsilon}{1+\epsilon}\right)^2 \cdot \mathcal{D}_{BW_N,\mathbf{c},\Sigma}(\mathbf{v}) \leq \mathfrak{D}(\mathbf{v}) \leq \left(\frac{1+\epsilon}{1-\epsilon}\right) \cdot \mathcal{D}_{BW_N,\mathbf{c},\Sigma}(\mathbf{v})$$

Since $\left(\frac{1-\epsilon}{1+\epsilon}\right)^2 \approx 1-4\epsilon$ and $\frac{1+\epsilon}{1+\epsilon} \approx 1+2\epsilon$, we have

$$(1-4\epsilon) \cdot \mathcal{D}_{BW_N,\mathbf{c},\Sigma}(\mathbf{v}) \leq \mathfrak{D}(\mathbf{v}) \leq (1+2\epsilon) \cdot \mathcal{D}_{BW_N,\mathbf{c},\Sigma}(\mathbf{v}).$$

From this inequality, the desired result can finally be obtained.

$$\sup \left| \frac{\mathfrak{D}(\mathbf{v})}{\mathcal{D}_{BW_N,\mathbf{c},\Sigma}(\mathbf{v})} - 1 \right| \leq 4\epsilon$$

Moreover, the statistical distance between the output distribution of the algorithm 7 and the discrete gaussian distribution over BW_N is equal 2ϵ. □

Smoothing parameter

A critical aspect in the design of a discrete Gaussian sampler for the Barnes–Wall lattice BW_N is determining the value of a suitable smoothing parameter. It is the smallest width of Gaussian noise that completely "smooths out" the discrete structure of the lattice. It allows us to sample from the lattice in a way that is indistinguishable from a uniform distribution which is crucial for the security of cryptographic schemes based on hard lattice problems such as SVP, CVP or, as in our case, LIP.

Definition 13 (Smoothing parameter in Euclidean norm) For a n-dimensional lattice Λ and a real value $\epsilon > 0$, we define its smoothing parameter $\eta_\epsilon(\Lambda)$ as the smallest value s such that $\rho_{1/s}(\Lambda^\star \backslash \{\mathbf{0}\}) \leq \epsilon$.

The iterative approach utilizes the sampler $\mathcal{D}_{(\theta^n BW_1)^{2^n},\mathbf{c},\Sigma}$. Consequently, we need to precisely determine the smoothing parameter $\eta_\epsilon(\theta^n BW_1)$ such that $\eta_\epsilon(\theta^n BW_1) \succ \Sigma$.

To address this, we use the exact formula for η_ϵ from [13]. For a lattice Λ and $\epsilon > 0$, we have

$$\eta_\epsilon(\Lambda) = \frac{1}{\lambda_1(\Lambda^\star)} \cdot \sqrt{\frac{1}{\pi} \ln \left(\frac{\kappa(\Lambda^\star)}{\epsilon} (1 + o(1)) \right)}.$$

Since our iterative sampler operates with the lattice $\theta^n BW_1$, we first prove the next proposition:

Proposition 9 *For $n \in \mathbb{N}$ we have:*

$$((1+i)^n BW_1)^\star = \frac{1}{(1-i)^n} BW_1.$$

Proof We know that BW_1 is an unimodular lattice, i.e. $BW_1^\star = BW_1$.

Due to $BW_1 = Z[i] \subset \mathbb{C}$, we can determine dual lattice $BW_1^\star$ as follows:

$$BW_1^\star = \{x \in \mathbb{C} | \langle x, y \rangle \in \mathbb{Z}[i] \text{ for all } y \in BW_1\},$$

where inner product determine as $\langle x, y \rangle = x \cdot \overline{y}$.

Let $x \in ((1+i)^n BW_1^\star)$ then for $y = (1+i)^n \cdot y_1 \in (1+i)^n BW_1$ and $y_1 \in BW_1$ we have

$$\langle x, y \rangle = x \cdot \overline{y} = x \cdot \overline{(1+i)^n \cdot y_1} = (1-i)^n \cdot x \cdot y_1 =< (1-i)^n \cdot x, y_1 >\in \mathbb{Z}[i]$$

Due to $y_1 \in BW_1$ it means that $(1-i)^n x \in BW_1^\star$. As a result $x \in ((1-i)^n)^{-1} BW_1^\star$ and $((1+i)BW_1)^\star \subset ((1-i)^n)^{-1} BW_1^\star$

The reverse inclusion is easy to prove in the same way. Finally, we have that $((1+i)BW_1)^\star = ((1-i)^n)^{-1} BW_1^\star$. Also, as we said previously $BW_1^\star = BW_1$. So it means that:

$$((1+i)^n BW_1)^\star = \frac{1}{(1-i)^n} BW_1^\star = \frac{1}{(1-i)^n} BW_1.$$

□

In other words, the dual of $(1+i)^n BW_1$ is its scaled version with scaling factor $\frac{1}{(1-i)^n}$.

Based on the above formulation, we now need to determine the following parameters for the dual lattice:

1. $\lambda_1(((1+i)^n BW_1)^\star) = \lambda_1 \left(\frac{1}{(1-i)^n} BW_1 \right) = \frac{1}{(\sqrt{2})^n} \lambda_1(BW_1) = \frac{1}{(\sqrt{2})^n} \lambda_1(\mathbb{Z}^2) = \frac{1}{(\sqrt{2})^n}$.
2. Kissing number $\tau\left((1+i)^n BW_1)^\star\right) = \tau \left(\frac{1}{(1-i)^n} BW_1 \right) = \tau(BW_1) = 4$. The kissing number remains invariant under lattice scaling [10].

As an example, we provide a table of smoothing parameter values for the lattices $\theta^n BW_1$ with $1 \leq n \leq 9$ and $\epsilon = (1/2)^2$.

n	$\eta_{(1/2)^2}(\theta^n BW_1)$
1	1.32856
2	1.87887
3	2.65713
4	3.75775
5	5.31426
6	7.51550
7	10.62852
8	15.03100
9	21.25704

5 Cryptanalysis of LIP

The LIP problem has recently been proposed as a foundation for cryptography in [2, 27], however, among all problems related to isometries, the LIP problem is the least studied. This section presents the current state of the art in LIP cryptanalysis, including all known attacks to date and reductions to the more well-known and long-studied problems of finding isometries for linear codes and graphs. In conclusion, we summarize the applicability of Barnes–Wall lattices and highlight the potential benefits of incorporating this family of lattices into cryptographic constructions based on the LIP problem

5.1 Related problems and reductions

The complexity of finding isometries for various algebraic structures has been studied for many years. Currently, there are works confirming the practical computational difficulty of solving the LIP problem [11, 28], as well as various code equivalence problems [29].

One of the most well-studied problems is the Graph Isomorphism (GI) problem. There exists a quasi-polynomial-time algorithm solving GI in time $\mathcal{O}(2^{(\log n)^{\mathcal{O}(1)}})$. Although this problem is not NP-complete, it is currently used to establish the computational hardness of other isomorphism problems.

The first attempt to prove GI-hardness of finding isometries was made for linear codes in [30]. The authors present a reduction from GI to the Permutation Code Equivalence (PCE) problem, thereby demonstrating that this problem is not NP-complete. Alternative reductions were proposed in [31, 32], but each had certain limitations, which were addressed in [33], where the authors demonstrate the reduction from GI to the Linear Code Equivalence (LCE) problem.

Theorem 10 ([33]) *For any field $\mathbb{F}$, the GI problem for graphs with n vertices and m edges reduces in polynomial time in n to the problem $\mathrm{LCE}_{\mathbb{F}}$ for codes of dimension $n+2$ and block length $m+3n$.*

The GI-hardness of the LIP problem was first proven in [34], but until recently, no reduction from coding theory problems to the LIP problem was known. This gap was also addressed in [33], where the authors show a reduction from the LCE problem to the LIP problem.

Theorem 11 ([33]) *Let p be a prime and $n, k \in \mathbb{Z}^+$. Then the* LCE_p *problem for $[n,k]_p$ codes reduces in time* $\mathrm{poly}(n,p)$ *to the* LIP *problem on lattices of rank R, where $R := 5n$ for $p = 2$ and $R := (p-1)(p+2)n = \left(p^2+p-2\right)n$ for $p > 2$.*

It is also worth noting that there is a known reduction from the Permutation Code Equivalence (PCE) problem to the GI problem for codes with trivial hull $\mathcal{C} \cap \mathcal{C}^\perp$ [35], but no such result has yet been presented for the LIP problem.

Finally, in preprint [36], authors proved equivalence between PCE and LCE problems. This result is significant for coding theory and for all cryptographic primitives based on the hardness of finding isometries, including LIP. The result, together with previously established reductions from [33], directly implies that PCE reduces to LIP in polynomial time. Despite several known reductions from code isometry problems to LIP, the converse reduction has not yet been presented in the literature. Thus, the equivalence between PCE and LIP remains an open question.

Thus, thanks to the presented results, a lower bound on the complexity of solving the LIP problem is known. Despite the existence of a quasipolynomial-time algorithm for solving the GI problem, no similar results have been presented for codes (LCE) and lattices (LIP) in the worst case (self-dual codes and unimodular lattices), and the most efficient attacks to date are based on finding low-weight codewords or short vectors in a lattice.

5.2 Attacks on LIP

In this section, we will study all known methods for solving the LIP problem, as well as cryptanalysis techniques for schemes based on this problem.

5.2.1 Graph-based approach

Let us introduce the notion of a characteristic set.

Definition 14 The function $\mathcal{V}$ defining a characteristic set is a mapping that assigns to each n and quadratic form $\mathbf{Q} \in \mathcal{S}_n^{>0}(\mathbb{R})$ a finite subset of vectors $\mathcal{V}(\mathbf{Q}) \subset \mathbb{Z}^n$ such that:

1. $\mathcal{V}(\mathbf{Q})$ generates $\mathbb{Z}^n$ (as a $\mathbb{Z}$-module);
2. For all $\mathbf{U} \in \mathcal{GL}_n(\mathbb{Z})$, we have $\mathbf{U}^{-1}\mathcal{V}(\mathbf{Q}) = \mathcal{V}(\mathbf{U}^T\mathbf{Q}\mathbf{U})$.

For characteristic sets $\mathcal{V}(\mathbf{Q})$ and their corresponding quadratic forms, the following lemma holds:

Lemma 12 ([5]) *Let $\mathcal{V}$ be a characteristic function, and let $\mathbf{Q}, \mathbf{Q}' \in \mathcal{S}_n^{>0}(\mathbb{R})$ be quadratic forms. Then there exists a bijection*

$$\mathrm{Isom}(\mathbf{Q}, \mathbf{Q}') \leftrightarrow \mathrm{Isom}(\mathcal{V}(\mathbf{Q}), \mathcal{V}(\mathbf{Q}')), \tag{1}$$

$$\mathbf{U} \leftrightarrow (\psi_{\mathbf{U}} : \mathbf{x} \to \mathbf{U}^{-1}\mathbf{x}), \tag{2}$$

where $\mathrm{Isom}(\mathcal{V}(\mathbf{Q}), \mathcal{V}(\mathbf{Q}'))$ *is the set of isometries between* $\mathcal{V}(\mathbf{Q})$ *and* $\mathcal{V}(\mathbf{Q}')$*. If* $\mathbf{Q} = \mathbf{Q}'$*, then* $\mathrm{Isom}(\mathbf{Q}, \mathbf{Q}') = \mathrm{Aut}(\mathbf{Q})$ *and* $\mathrm{Isom}(\mathcal{V}(\mathbf{Q}), \mathcal{V}(\mathbf{Q}'))$ *are groups under composition, and the bijection is an isomorphism.*

Thus, by finding an isometry between two characteristic sets, we can conclude the isomorphism of the corresponding quadratic forms.

The general approach [37] to finding such isometries consists of constructing a weighted graph for $\mathbf{Q}$ and $\mathbf{Q}'$ respectively, where the vertices are the elements of the characteristic set $\mathcal{V}(\mathbf{Q})$ (or $\mathcal{V}(\mathbf{Q}')$, respectively), and an edge $(\mathbf{x}, \mathbf{y})$ has weight $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{Q}}$ or $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{Q}'}$, respectively.

This representation allows the equivalence problem of quadratic forms to be reduced to the graph isomorphism problem. Lemma 162 from [5] states that two forms are equivalent if and only if their corresponding graphs are isomorphic.

There are various algorithms to solve the graph isomorphism problem, for example, Nauty, Traces [7], Bliss [8]. These algorithms explore different vertex mappings that preserve edge weights. A critical factor for such algorithms is the size of the characteristic set $|\mathcal{V}(\mathbf{Q})|$. For instance, the Babai algorithm [9], which asymptotically works in quasipolynomial time in $|\mathcal{V}(\mathbf{Q})|$, may in practice operate in $\exp(n^{O(1)})$ time, since the size $|\mathcal{V}(\mathbf{Q})|$ is of the order of $\exp(\Theta(n))$.

5.2.2 The Haviv-Regev algorithm [28]

The Haviv-Regev algorithm [28] runs in time $n^{O(n)}$ and returns all solutions to the LIP problem. This solution is not always optimal, as the size of the automorphism group for certain lattices can be quite large. The main open question is whether there exists a single algorithm that finds one solution to the LIP problem.

Definition 15 For the set $C \subseteq \mathbb{Z}^n$, a *chain of length* d in C is a sequence of d vectors $x_1, \ldots, x_d \in C$ such that $x_j \notin \mathrm{span}(\{x_1, \ldots, x_{j-1}\})$ for every $2 \le j \le d$. If, in addition, $C \subseteq \mathrm{span}(\{x_1, \ldots, x_d\})$ we say that the chain is *maximal.* We say that a vector $z \in \mathbb{Z}^n$ *uniquely defines* a chain $x_1, \ldots, x_d$ in C if for every $1 \le j \le d$, the minimum inner product of z with vectors in $C \setminus \mathrm{span}(\{x_1, \ldots, x_{j-1}\})$ is uniquely achieved by x_j.

Theorem 13 ([28, Theorem 4.1]) *There exists an algorithm that given two bases of lattices* Λ_1 *and* Λ_2 *of rank* n*, outputs all orthogonal linear transformations* $O : \mathrm{span}(\Lambda_1) \to \mathrm{span}(\Lambda_2)$ *for which* $\Lambda_2 = O(\Lambda_1)$ *in running time* $n^{O(n)} \cdot s^{O(1)}$ *and in polynomial space, where* s *denotes the input size. In addition, the number of these transformations is at most* $n^{O(n)}$*.*

Let Λ_1 and Λ_2 be the lattices with associated bases B_1 and B_2. For $i \in \{1, 2\}$, the Algorithm 8 computes the set A_i of all shortest nonzero vectors of Λ_i and the set W_i of all vectors in the dual lattice $\Lambda_i^\star$ of norm at most $5n^{17/2} \cdot \lambda_1(\Lambda_i^\star)$. Given these sets, the algorithm finds a $w_1 \in W_1$ that uniquely defines a linearly independent chain of length n in A_1 and the corresponding chain $x_1, \ldots, x_n \in A_1$. The existence of w_1 is guaranteed by the following theorem.

Theorem 14 ([28, Theorem 4.2]) *Let Λ be a lattice of rank n satisfying $\lambda_1(\Lambda) = \lambda_n(\Lambda)$, and let A denote the set of all shortest nonzero vectors of Λ. Then there exists a vector $v \in \Lambda^\star$ that uniquely defines a linearly independent chain of length n in A and satisfies*

$$\|v\| \leq 5n^{17/2} \cdot \lambda_1(\Lambda^\star).$$

Then, the algorithm goes over all vectors $w_2 \in W_2$ and for every w_2 which uniquely defines a linearly independent chain $y_1, \ldots, y_n \in A_2$ it checks if the linear transformation $O : \mathrm{span}(\Lambda_1) \to \mathrm{span}(\Lambda_2)$, defined by $O(x_i) = y_i$ for every $1 \leq i \leq n$, is orthogonal and maps Λ_1 to Λ_2. If this is the case, then O is inserted to the output set.

Algorithm 8 Lattice Isomorphism – Special Case [28]

Require: Two bases of lattices Λ_1 and Λ_2 of rank n satisfying $\lambda_1(\Lambda_1) = \lambda_n(\Lambda_1)$ and $\lambda_1(\Lambda_2) = \lambda_n(\Lambda_2)$.

Ensure: The set *Output* of all orthogonal linear transformations $O : \mathrm{span}(\Lambda_1) \to \mathrm{span}(\Lambda_2)$ for which $\Lambda_2 = O(\Lambda_1)$.

1: **for all** $i = 1, 2$ **do**
2:     $A_i \leftarrow \{x \in \Lambda_i \mid \|x\| = \lambda_1(\Lambda_i)\}$
3:     $W_i \leftarrow \{x \in \Lambda_i^\star \mid \|x\| \leq 5n^{17/2} \cdot \lambda_1(\Lambda_i^\star)\}$
4:     Corollary 2.16
5: **end for**
6: **for all** $w_1 \in W_1$ **do**
7:     **if** w_1 uniquely defines a linearly independent chain of length n in A_1 **then**
8:         $(x_1, \ldots, x_n) \leftarrow$ the maximal linearly independent chain that w_1 uniquely defines in A_1
9:         **goto** line 11
10:     **end if**
11: **end for**
12: **for all** $w_2 \in W_2$ **do**
13:     **if** w_2 uniquely defines a linearly independent chain of length n in A_2 **then**
14:         $(y_1, \ldots, y_n) \leftarrow$ the maximal linearly independent chain that w_2 uniquely defines in A_2
15:         $O \leftarrow$ the linear transformation that maps x_i to y_i for every $1 \leq i \leq n$
16:         **if** O is orthogonal and satisfies $\Lambda_2 = O(\Lambda_1)$ **then**
17:             $\mathit{Output} \leftarrow \mathit{Output} \cup \{O\}$
18:         **end if**
19:     **end if**
20: **end for**

This algorithm provides an asymptotic bound for solving the LIP problem; however, the proven constant in the exponent is rather large.

5.2.3 Hull attack

The paper [11] investigates the possibility of using s-hulls of lattices for attacks on the LIP problem. In particular, for certain lattices, the minimal distance of the hull is relatively smaller than that of the original lattice, which can consequently be used to improve the efficiency of algorithms solving the LIP problem. Note, however, that the cost of attacks based on this approach remains exponential, but the constant in the exponent is reduced by half. Therefore, when constructing cryptographic algorithms, the proper choice of lattice is an important consideration.

The works [2, 11] identify lattices that are of particular interest for constructing schemes based on the LIP problem – unimodular lattices characterized by the property $\Lambda = \Lambda^\star$. In particular, unimodular lattices include $\mathbb{Z}^n$, which serves as the foundation for the digital signature scheme Hawk [38], as well as Barnes–Wall lattices which we consider in this work.

Definition 16 (Hull of a lattice) For a lattice Λ and its dual lattice $\Lambda^\star$, the hull is defined as follows:

$$H(\Lambda) = \Lambda \cap \Lambda^\star.$$

Note that $\Lambda \subseteq \Lambda^\star$, so when defining the classical hull, we always obtain $H(\Lambda) = \Lambda$. Because of this, the generalized version of this construction – the s-hull – is of greater research interest.

Definition 17 (s-hull) For $s \in \mathbb{R}^\times$, the s-hull of lattice Λ is defined as the sublattice

$$H_s(\Lambda) = \Lambda \cap s\Lambda^\star.$$

In work [11], the authors present Lemma 1, which characterizes the structures of s-hulls depending on s. If Λ is integral lattice, then the following holds:

1. If s is irrational, then $H_s(\Lambda) = \{0\}$.
2. If $s = a/b \in \mathbb{Q}$, then $H_s(\Lambda) = H_a(\Lambda) = aH(\Lambda) = a\Lambda$.
3. If $s \in \mathbb{Z}$, then $H_s(\Lambda) = sH(\Lambda) = s\Lambda$.

Thus, we observe that in the case of unimodular lattices, the s-hull can either be trivial (for irrational s) or a scaled version of the original lattice.

Note also that Lemma 1 eliminates the need to consider an infinite number of s values. It is sufficient to consider only those s that divide $\det \mathbf{Q}$. All other s-hulls will be scaled versions of t-hulls for some $t | \det \mathbf{Q}$. Moreover, it is known that the gap value for lattice Λ equals the gap value for $s\Lambda$. Therefore, the complexity of attacks on the LIP problem for lattices Λ and $s\Lambda$ is the same. Consequently, when considering a unimodular lattice, we conclude that the only s-hull worth examining according to Lemma 1 [11] is $H_1(\Lambda) = \Lambda$. However, in this case, it is obvious that attacks on the LIP problem are not simplified, as the attacker is left with only the original lattice.

The Barnes–Wall lattices are defined over the ring of Gaussian integers $\mathbb{Z}[i]$, so the results of the work are not directly applicable to them. In [11] authors show that half the Barnes–Wall lattices are unimodular up to scaling:

$$BW_N = \begin{bmatrix} 1 & 0 \\ 1 & 1+i \end{bmatrix}^{\otimes n}, \quad \mathbf{Q}_N = \left(\mathbf{B}_1^\dagger \cdot \mathbf{B}_1\right)^{\otimes n} = \begin{bmatrix} 2 & 1-i \\ 1+i & 2 \end{bmatrix}^{\otimes n},$$

where $N = 2^n$ and $\mathbf{B}_1^\dagger$ is the conjugate transpose of the matrix $\mathbf{B}$.

For $n = 2$ we can show that lattice and associated quadratic forms are unimodular:

$$\mathbf{Q}_4' = \frac{1}{2} \cdot \mathbf{Q}_4 = \begin{bmatrix} 2 & 1+i & 1+i & i \\ 1-i & 2 & 1 & 1+i \\ 1-i & 1 & 2 & 1+i \\ -i & 1-i & 1-i & 2 \end{bmatrix}, \text{ with } \det(\mathbf{Q}_4') = 1.$$

Since $\mathbf{Q}_{4^n} = \mathbf{Q}_4^{\otimes n}$, each associated $\mathbf{Q}'_{4^n} = \mathbf{Q}_{4^n}/2^n$ has determinant 1 over $\mathbb{Z}[i]$. Consequently, the Barnes–Wall lattices of rank $N = 2^{2n}$ are unimodular over $\mathbb{Z}[i]$ (up to scaling). Moreover, every Barnes–Wall lattice with rank $N = 2^n$ is unimodular over the field of complex numbers $\mathbb{C}$:

$$\mathbf{Q}_2' = \frac{1}{\sqrt{2}} \mathbf{Q}_2 = \begin{bmatrix} \sqrt{2} & \frac{1-i}{\sqrt{2}} \\ \frac{1+i}{\sqrt{2}} & \sqrt{2} \end{bmatrix}, \text{ with } \det(\mathbf{Q}_2') = 1.$$

So, we have that $\mathbf{Q}_{2^n} = \mathbf{Q}_2^{\otimes n}$ and quadratic forms $\mathbf{Q}'_{2^n} = \mathbf{Q}_{2^n}/\sqrt{2^n}$ have determinant 1 over $\mathbb{Z}[i]$, which proves the self-duality over $\mathbb{C}$.

Our computational experiments demonstrate that the dimension of s-hulls of Barnes–Wall lattices, defined over Gaussian integers, matches the dimension of the original lattice for any s:

$$H_s(\Lambda) = \mathbf{B} \cdot \Lambda_s^\star(\mathbf{B}^\top \mathbf{B}) = \mathbf{B} \cdot \left\{ \mathbf{x} \in \mathbb{Z}[i]^n : \mathbf{B}^\top \mathbf{B} \mathbf{x} = 0 \mod s \right\}, \text{ with } \dim(H_s(\Lambda)) = N,$$

where $\Lambda_q^\star(\mathbf{A}) = \{\mathbf{x} \in \mathbb{Z}[i]^n : \mathbf{A}\mathbf{x} = 0 \mod q\}$.

Remark 2 As shown in Section 3.3, when justifying the inapplicability of attacks related to s-hulls to Barnes–Wall lattices, one can transition to integer versions of the lattices. Note that in this case, the results from [11] can be applied, since their results hold for any integer lattices, unlike the Gaussian integers case.

Therefore, computing s-hulls does not provide an adversary with any additional advantage in the cryptanalysis of LIP in this context, and thus does not pose a threat to the security of the schemes based on this problem.

5.3 Coding Theory attacks

It is known that algebraic lattices are closely related to error-correcting codes. The analogs of the lattice isomorphism problem in coding theory are the problems of *Permutation Code Equivalence* (PCE) and *Linear Code Equivalence* (LCE). Both the complexity of these problems and the methods for solving them have been studied for a much longer than those for the LIP problem. Therefore, it is necessary to take a closer look at approaches and attacks from coding theory, as well as their potential adaptation to the lattice setting.

Let $\mathrm{Mono}_n(\mathbb{F}_q)$ denote the set of monomial matrices of the form $\mathbf{M} = \mathbf{DP}$, where $\mathbf{D} \in \mathbb{F}_q^{n \times n}$ is a full-rank diagonal matrix and $\mathbf{P} \in \mathrm{Perm}_n(\mathbb{F}_q)$.

Definition 18 (Linear Code Equivalence) Let $\mathbf{G}, \mathbf{G}' \in \mathbb{F}_q^{k \times n}$ be generator matrices of $[n, k]$-codes $\mathcal{C}, \mathcal{C}'$ respectively. The codes $\mathcal{C}, \mathcal{C}'$ are linearly equivalent if there exist $\mathbf{S} \in \mathcal{GL}_k(\mathbb{F}_q)$ and $\mathbf{Q} \in \mathrm{Mono}_n(\mathbb{F}_q)$ such that $\mathbf{G}' = \mathbf{SGQ}$.

Solving the LCE problem means determining whether such matrices $\mathbf{S}, \mathbf{Q}$ exist for two given linear codes. Note that LCE generalizes PCE to the case of isometries that are not necessarily permutations. If $\mathbf{Q} \in \mathrm{Perm}_n(\mathbb{F}_q)$, then the codes are permutation-equivalent.

In [39], it was shown that the LCE problem can be reduced to the PCE problem. Currently, there are two main classes of algorithms for solving code equivalence problems. These will be considered in more detail below.

Invariant-based attacks

In the work of Sendrier [40], the Support Splitting Algorithm (SSA) is introduced, which solves the PCE problem over $\mathbb{F}_2$. This algorithm determines in polynomial time whether two codes are permutation-equivalent, i.e., whether a permutation of the columns of one generator matrix yields the other.

Theorem 15 (SSA Complexity) *Let $\mathcal{C}$ and $\mathcal{C}'$ be linear $[n, k]$ codes over $\mathbb{F}_2$. Then the SSA algorithm returns in time $O\left(n^3 + 2^h n\ell(n)\right)$ a permutation $\sigma \in \mathcal{S}_n$ (if it exists) such that $\sigma(\mathcal{C}) = \mathcal{C}'$, where $h = \dim\left(\mathcal{C} \cap \mathcal{C}^\perp\right)$ and $\ell(n)$ is a parameter of the SSA algorithm.*

Sendrier assumes $\ell(n) = \ln(n)$. Moreover, according to [41], the parameter h is very small for most linear codes, including Goppa codes. This implies that the practical complexity of the algorithm over $\mathbb{F}_2$ is polynomial in the code length.

To solve the LCE problem, the SSA algorithm was generalized in [39] to codes over $\mathbb{F}_q$ for $q \in \{3, 4\}$. The original version of the algorithm used the codeword weight distribution as an invariant, which is easy to compute when the code dimension is small. The authors use the hull of the code $R = \mathcal{C} \cap \mathcal{C}^\perp$ for this purpose. However, this approach does not apply when $q > 5$, as the hull is not invariant under general isometries. As an alternative, the authors proposed using of the *closure of the code* $\{c \otimes \mathbf{a} \mid c \in \mathcal{C}\}$, where $\mathbf{a} = (a_1, \ldots, a_{q-1})$ and a_i are distinct elements of $\mathbb{F}_q^*$, to construct new invariants. However, this solution was effective only for $q < 5$.

Although there exists a polynomial-time algorithm for solving the PCE problem for binary codes, the complexity of SSA and its modifications strongly depends on the dimension of the code's hull $\dim\left(\mathcal{C} \cap \mathcal{C}^{\perp}\right)$. This dimension is maximal in the case of self-dual codes.

Field	Random Codes	Self-dual Codes (Worst Case)
$\mathbb{F}_2$	$\mathcal{O}(n^3)$	$\mathcal{O}(2^k n^2 \log n)$
$\mathbb{F}_3$	$\mathcal{O}(n^3)$	$\mathcal{O}(3^k n^2 \log n)$
$\mathbb{F}_4$	$\mathcal{O}(n^3)$	$\mathcal{O}(2^{2k} n^2 \log n)$
$\mathbb{F}_q$, $q \geq 5$	$\mathcal{O}(q^k n^2 \log n)$	$\mathcal{O}(q^k n^2 \log n)$

Table 2 SSA Algorithm Complexity

Finding efficiently computable invariants for code equivalence remains an open problem, and despite many years of research, no significantly improved variants of SSA have been proposed.

In the preprint [6], the authors study how the Schur product can be employed as a tool to improve known solvers for the code equivalence problem. They demonstrate that if the square of two codes $\mathcal{A}, \mathcal{B}$ does not span the entire space $\mathbb{F}_q^n$, then the dimension of the hull of their squares is small in most cases. This result establishes that the PCE problem can be solved for any codes when the dimensions of the considered codes are either less than $\sqrt{2n}$ or greater than $n - \sqrt{2n}$. Consequently, this allows efficient solutions to both PCE and LCE problems for codes with rates approaching either 0 or 1, while the dimension of the hull of the original codes may be arbitrary. Furthermore, the work demonstrates how to apply the Schur square to efficiently solve the PCE problem through puncturing and computing the square of the hull, provided that the hull dimension is less than $\sqrt{2n}$. Notably, these results hold for codes with arbitrary parameters. Despite significant recent progress in the cryptanalysis of PCE, solving it in the worst-case scenario – particularly for self-dual codes – remains a challenging task.

It is worth noting that in [42], the authors presented a polynomial-time algorithm for recovering a secret isomorphism when it is reused to generate a public key in schemes relying on the hardness of the t-LCE problem. However, this result does not affect the security of framework [2].

Low-weight codeword search attacks

The fastest algorithms for solving PCE and LCE in the worst case are based on searching for codewords with low Hamming weight (or subcodes with small supports) [29, 43, 44]. These attacks share a common principle, which we now detail.

In Leon's algorithm [43], all codewords of weight $\mathrm{wt}(c) \leq w$ are found for both codes:

$$A = \{\mathbf{c} \in \mathcal{C} \mid \mathrm{wt}(\mathbf{c}) \leq w\}, \quad A' = \{\mathbf{c}' \in \mathcal{C}' \mid \mathrm{wt}(\mathbf{c}') \leq w\}.$$

It follows that $A\mathbf{Q} = A'$, and when $w \ll n$, we have $|A| \ll |\mathcal{C}| = q^k$. Since A and A' are small, recovering $\mathbf{Q}$ becomes easier. Modern algorithms relax the constraints of [43] and instead aim to find many intersections, i.e., pairs $(\mathbf{c}, \mathbf{c}')$ such that $\mathbf{c}\mathbf{Q} = \mathbf{c}'$. This idea was first proposed in [44] and later improved in [29]. The approaches of [29, 44] outperform Leon's algorithm only when q is sufficiently large.

Thus, the attack process involves the following steps:

1. Construct two sets L_1 and L_2 containing low-weight codewords with at least X overlaps.
2. Find the intersection of L_1 and L_2.
3. Use the intersection to recover the secret matrix $\mathbf{Q}$.

It is known that the first step is far more complex than computing the intersection or recovering $\mathbf{Q}$. The authors of [45] propose the following worst-case parameter selection criterion:

$$\frac{1}{\sqrt{N(w)}} \cdot C_{\mathrm{ISD}}(w) > 2^{\lambda}, \quad N(w) = \binom{n}{w}(q-1)^{w} q^{-(n-k)}$$

for any w, where $q \geq 5$, w is an upper bound on codeword weight, and $C_{\mathrm{ISD}}(\cdot)$ is the complexity of the ISD algorithm.

The lattice-based analogs of these attacks involve finding short lattice vectors. As noted earlier, the PCE and LCE problems can be reduced to LIP, so lattice-based attacks may be used to find low-weight codewords (see [33]). However, the reverse direction is currently unknown. Thus, the cryptanalysis of the PCE and LCE problems is more extensive compared to that of the LIP problem. Moreover, the results of [6] suggest that, at present, secure schemes based on the PCE problem can only be constructed using self-dual codes or codes with a large hull; however, constructing such codes is a highly nontrivial task. It should be noted that, due to the structural similarities between codes and lattices, these attacks may potentially be generalized to the LIP problem, although no such results are currently known. In any case, by analogy with self-dual codes, the use of self-dual lattices, according to [11], appears to be the most secure and conservative approach when designing schemes based on the LIP problem. As noted, Barnes–Wall lattices are self-dual up to scaling, which makes them highly promising candidates for building cryptographic schemes.

5.4 Conclusions on LIP hardness with Barnes–Wall lattices

All currently known methods of cryptanalysis of the LIP problem reduce to finding short vectors in a lattice. Thus, the complexity of LIP can be estimated using the complexity of solving SVP or uSVP problem, depending on the particular scheme. The most efficient known algorithm for finding short vectors in lattices is the Block Korkine–Zolotarev (BKZ) algorithm [46, 47]. Its main idea is to iteratively solve the Shortest Vector Problem (SVP) in sublattices of dimension β, called the block size. The cost of BKZ-β is dominated by calls to an SVP oracle in dimension β, and can be expressed as

$$\mathrm{poly}(\beta) \cdot 2^{c\beta + o(\beta)},$$

where the exponential term corresponds to the complexity of solving SVP in dimension β. In cryptographic applications, the *core-SVP paradigm* is used to estimate the hardness of BKZ. This heuristic disregards polynomial factors and lower-order terms, considering only the main exponential contribution $2^{c\beta}$. While this underestimates the true running time, it provides a conservative and widely adopted measure of security.

Definition 19 (Gaussian Heuristic) Let $\mathcal{L}$ be a lattice of rank n with volume $\mathrm{vol}(\mathcal{L})$. The expected value $\mathrm{gh}(\mathcal{L})$ of the first minimum $\lambda_1(\mathcal{L})$ according to the Gaussian heuristic is:

$$\mathrm{gh}(\mathcal{L}) = \frac{\mathrm{vol}_n(\mathcal{L})^{1/n}}{\mathrm{vol}(\mathcal{B}_1^n)^{1/n}} \approx \sqrt{\frac{n}{2\pi e}} \cdot \mathrm{vol}(\mathcal{L})^{1/n}.$$

The behavior of Gram–Schmidt vectors under BKZ reduction is commonly predicted by the *Geometric Series Assumption (GSA)*. It states [48] that, for a BKZ-β reduced basis, the Gram–Schmidt vector norms decrease geometrically with ratio $\alpha_\beta = \mathrm{gh}(\beta)^{2/(\beta-1)}$ and the norm of the shortest vector in BKZ-β reduced basis satisfies:

$$\|\mathbf{b}_0\| \lessapprox \sqrt{\alpha_\beta}^{\,n-1} \cdot \mathrm{vol}(\mathcal{L}(\mathbf{B}))^{\frac{1}{n}}.$$

Alkim et al. [49] derived a condition for recovering unusually short vectors in a lattice. In particular, if $\mathbf{B}$ is BKZ-β reduced, then a target vector v can be found whenever

$$\sqrt{\frac{\beta}{n}} \cdot \|v\| \; < \; \delta_\beta^{2\beta-n-1} \cdot \mathrm{vol}(\Lambda(\mathbf{B}))^{1/n}, \tag{3}$$

where $\delta_\beta = (\|\mathbf{b}_0\|/\mathrm{vol}(\Lambda)^{1/n})^{1/n} \approx \sqrt{\alpha_\beta}$ is root-Hermite factor.

This inequality provides the link between block size β, lattice parameters, and security estimates. The best known classical algorithm [50] for solving SVP achieves $c_C = \log_2 \sqrt{3/2} \approx 0.292$, while the best quantum algorithm [51] achieves $c_Q = \log_2 \sqrt{13/9} \approx 0.265$. Even with further improvements, the complexity of a quantum algorithm for solving SVP is unlikely to be smaller than $\sqrt{4/3}^{\beta+o(\beta)} \approx 2^{0.2075\beta}$.

N'	**Estimated β**	**Classical Security**	**Quantum Security**
512	426	124	112
1024	816	238	216
2048	1584	462	419

Table 3 Security estimates for $BW_{N'}^{\mathbb{Z}}$

Based on the evaluation of the required block size in 3, the Shortest Vector Problem (SVP) in the Barnes–Wall lattice was analyzed, incorporating complexity estimates of SVP-solving algorithms for both classical and quantum scenarios. The estimates are displayed in Table 3. Security estimates for various dimensions of Barnes–Wall lattices were derived, where classical and quantum security, denoted by λ, corresponds to the complexity of solving SVP, expressed as 2^λ, for classical and quantum algorithms, respectively. To achieve more precise parameter estimates, it is recommended to employ specialized estimators, while also considering the parameters of the cryptographic scheme in which Barnes–Wall lattices are utilized.

Note that all the previous arguments, including those concerning the Gaussian heuristic, the norms of vectors in BKZ-β reduced bases, and the results of [49] on the hardness of uSVP, are valid for random lattices with $\lambda_1 \approx \mathrm{gh}(\Lambda)$. The Barnes–Wall lattices are not random, and the complexity of SVP in structured lattices remains an open question. Nevertheless, due to the lack of alternative estimates, when assessing the security of schemes on remarkable lattices, one typically relies on estimates for random lattices [3, 4, 52]. Despite this drawback, in the context of the LIP problem, Barnes–Wall lattices are extremely promising, since their use prevents the application of hull-based cryptanalysis methods, which were used, for example, in [6] to break the lattice-based UPKE scheme [14], whose security relies on the LWE and PCE problems. Moreover, Barnes–Wall lattices have high density and, as noted in Section 3.1, their gap is bounded by $\Theta(\sqrt[4]{n})$, which allows to select smaller parameters while achieving the same security level as for $\mathbb{Z}^n$ lattices. This potentially increases the efficiency of cryptographic schemes constructed on these lattices.

6 Conclusion

In this work, we have examined the application of Barnes–Wall lattices in the design of post-quantum lattice-based cryptographic schemes. We contend that Barnes–Wall lattices are particularly advantageous for cryptographic schemes based on the LIP due to the following reasons.

Firstly, Barnes–Wall lattices exhibit a higher density compared to, for instance, $\mathbb{Z}^n$ lattices, which increases the computational complexity of solving the Shortest Vector Problem. Additionally, it is noteworthy that the Hull attack does not reduce the complexity of SVP-solving algorithms due to the unimodularity of Barnes–Wall lattices. These two factors suggest that employing Barnes–Wall lattices enables the use of smaller system parameters, such as lattice dimension, while maintaining the required level of security.

Secondly, we propose two efficient Gaussian sampling algorithms for Barnes–Wall lattices, leveraging their specific construction structure. This novel algorithms facilitate more efficient sampling from a Gaussian distribution over Barnes–Wall lattices compared to existing general-purpose algorithms. The proposed algorithms enable the development of efficient digital signature schemes utilizing Barnes–Wall lattices.

A significant open question remains regarding the precise assessment of the complexity of solving SVP for Barnes–Wall lattices, given their unique structure. As current complexity estimation methods rely on heuristics originally developed for random lattices, further research is necessary to accurately evaluate the SVP complexity for Barnes–Wall lattices.

Statements and Declarations

Competing interests

The authors have no Conflict of interest to declare that are relevant to the content of this article This research received no external funding.

References

[1] Ducas, L., Pierrot, C.: Polynomial time bounded distance decoding near minkowski's bound in discrete logarithm lattices. Designs, Codes and Cryptography **87** (2019)

[2] Ducas, L., Woerden, W.: On the lattice isomorphism problem, quadratic forms, remarkable lattices, and cryptography. In: Dunkelman, O., Dziembowski, S. (eds.) Advances in Cryptology – EUROCRYPT 2022, pp. 643–673. Springer, Cham (2022)

[3] Castro Biage, G., Zambonin, G., Idalino, T.B., Panario, D., Custodio, R.: A concrete lip-based kem with simple lattices. IEEE Access **12**, 16408–16420 (2024) https://doi.org/10.1109/ACCESS.2024.3358670

[4] Branco, P., Malavolta, G., Maradni, Z.: Fully-Homomorphic Encryption from Lattice Isomorphism. Cryptology ePrint Archive, Paper 2025/993 (2025). https://eprint.iacr.org/2025/993

[5] Woerden, W.P.J.: Lattice cryptography, from cryptanalysis to new foundations. Phd thesis, Leiden University (2023). https://scholarlypublications.universiteitleiden.nl/access/item%3A3564772/download

[6] Battagliola, M., Mora, R., Santini, P.: Using the Schur Product to Solve the Code Equivalence Problem. Cryptology ePrint Archive, Paper 2025/1017 (2025). https://eprint.iacr.org/2025/1017

[7] McKay, B.D., Piperno, A.: Practical graph isomorphism, ii. Journal of symbolic computation **60**, 94–112 (2014)

[8] Junttila, T., Kaski, P.: Engineering an efficient canonical labeling tool for large and sparse graphs. In: 2007 Proceedings of the Ninth Workshop on Algorithm Engineering and Experiments (ALENEX), pp. 135–149 (2007). SIAM

[9] Babai, L.: Graph isomorphism in quasipolynomial time. In: Proceedings of the Forty-eighth Annual ACM Symposium on Theory of Computing, pp. 684–697 (2016)

[10] Conway, J., Sloane, N.: Sphere Packings, Lattices and Groups vol. 290, (1988). https://doi.org/10.1007/978-1-4757-2016-7

[11] Ducas, L., Gibbons, S.: Hull attacks on the lattice isomorphism problem. In: Boldyreva, A., Kolesnikov, V. (eds.) Public-Key Cryptography – PKC 2023, pp. 177–204. Springer, Cham (2023)

[12] Forney, G.D.: Coset codes. ii. binary lattices and related codes. IEEE Transactions on Information Theory **34**(5), 1152–1187 (1988) https://doi.org/10.1109/18.21246

[13] Espitau, T., Wallet, A., Yu, Y.: On Gaussian sampling, smoothing parameter and application to signatures. Cryptology ePrint Archive, Paper 2023/1654 (2023). https://eprint.iacr.org/2023/1654

[14] Albrecht, M., Benčina, B., Lai, R.W.F.: Hollow LWE: A New Spin: Unbounded Updatable Encryption from LWE and PCE, pp. 363–392 (2025). https://doi.org/10.1007/978-3-031-91101-9_13

[15] Corlay, V.: Decoding algorithms for lattices. PhD thesis, Institut Polytechnique de Paris (2020)

[16] Bollauf, M.F., Lie, M., Ling, C.: On Gaussian Sampling for q-ary Lattices and Linear Codes with Lee Weight. Cryptology ePrint Archive, Paper 2025/087 (2025). https://eprint.iacr.org/2025/087

[17] Berlekamp, E., McEliece, R., Tilborg, H.: On the inherent intractability of certain coding problems (corresp.). IEEE Transactions on Information Theory **24**(3), 384–386 (1978) https://doi.org/10.1109/TIT.1978.1055873

[18] Weger, V., Khathuria, K., Horlemann, A.-L., Battaglioni, M., Santini, P., Persichetti, E.: On the hardness of the lee syndrome decoding problem. Advances in Mathematics of Communications **18** (2022) https://doi.org/10.3934/amc.2022029

[19] Micciancio, D., Regev, O.: Worst-case to average-case reductions based on gaussian measures, vol. 37, pp. 372–381 (2004)

[20] Peikert, C., Rosen, A.: Efficient collision-resistant hashing from worst-case assumptions on cyclic lattices. In: Halevi, S., Rabin, T. (eds.) Theory of Cryptography, pp. 145–166. Springer, Berlin, Heidelberg (2006)

[21] Brakerski, Z., Langlois, A., Peikert, C., Regev, O., Stehlé, D.: Classical hardness of learning with errors. In: Proceedings of the Forty-Fifth Annual ACM Symposium on Theory of Computing. STOC '13, pp. 575–584. Association for Computing Machinery, New York, NY, USA (2013). https://doi.org/10.1145/2488608.2488680 . https://doi.org/10.1145/2488608.2488680

[22] Micciancio, D., Goldwasser, S.: Complexity of Lattice Problems: A Cryptographic Perspective vol. 671, (2002). https://doi.org/10.1007/978-1-4615-0897-7

[23] Babai, L.: On lovász' lattice reduction and the nearest lattice point problem. Combinatorica **6**(1), 1–13 (1986) https://doi.org/10.1007/BF02579403

[24] Haviv, I., Regev, O.: On the lattice isomorphism problem. Proceedings of the Annual ACM-SIAM Symposium on Discrete Algorithms (2013) https://doi.org/10.1137/1.9781611973402.29

[25] Klein, P.: Finding the closest lattice vector when it's unusually close. In: Proceedings of the Eleventh Annual ACM-SIAM Symposium on Discrete Algorithms, pp. 937–941 (2000)

[26] Gentry, C., Peikert, C., Vaikuntanathan, V.: Trapdoors for hard lattices and new cryptographic constructions, vol. 14, pp. 197–206 (2008)

[27] Bennett, H., Ganju, A., Peetathawatchai, P., Stephens-Davidowitz, N.: Just how hard are rotations of $\mathbb{Z}^n$? algorithms and cryptography with the simplest lattice. In: Hazay, C., Stam, M. (eds.) Advances in Cryptology – EUROCRYPT 2023, pp. 252–281. Springer, Cham (2023)

[28] Haviv, I., Regev, O.: On the lattice isomorphism problem. Proceedings of the Annual ACM-SIAM Symposium on Discrete Algorithms, 391–404 (2014) https://doi.org/10.1137/1.9781611973402.29

[29] Barenghi, A., Biasse, J.-F., Persichetti, E., Santini, P.: On the computational hardness of the code equivalence problem in cryptography. Advances in Mathematics of Communications **17** (2022) https://doi.org/10.3934/amc.2022064

[30] Petrank, E., Roth, R.M.: Is code equivalence easy to decide? IEEE Transactions on Information Theory **43**(5), 1602–1604 (1997) https://doi.org/10.1109/18.623157

[31] Kaski, P., Ostergard, P.: Classification Algorithms for Codes and Designs, (2005). https://doi.org/10.1007/3-540-28991-7

[32] Grochow, J.A.: Matrix isomorphism of matrix lie algebras. In: 2012 IEEE 27th Conference on Computational Complexity, pp. 203–213 (2012). https://doi.org/10.1109/CCC.2012.34

[33] Bennett, H., Win, K.M.H.: Relating code equivalence to other isomorphism problems. Designs, Codes and Cryptography (2024) https://doi.org/10.1007/s10623-024-01542-3

[34] Sikiric, M., Schuermann, A., Vallentin, F.: Complexity and algorithms for computing voronoi cells of lattices. Mathematics of Computation **78** (2008) https://doi.org/10.1090/S0025-5718-09-02224-8

[35] Bardet, M., Otmani, A., Saeed-Taha, M.: Permutation code equivalence is not harder than graph isomorphism when hulls are trivial. In: 2019 IEEE International Symposium on Information Theory (ISIT), pp. 2464–2468 (2019). https://doi.org/10.1109/ISIT.2019.8849855

[36] Cheraghchi, M., Shagrithaya, N., Veliche, A.: Reductions Between Code Equivalence Problems. Cryptology ePrint Archive, Paper 2025/208 (2025). https://eprint.iacr.org/2025/208

[37] Dutour Sikirić, M., Haensch, A., Voight, J., Woerden, W.P.: A canonical form for positive definite matrices. Open Book Series **4**(1), 179–195 (2020)

[38] Huang, T.P., Postlethwaite, E.W., Prest, T., Pulles, L.N., Woerden, W.: https://hawk-sign. info

[39] Sendrier, N., Simos, D.: How easy is code equivalence over fq? (2013)

[40] Sendrier, N.: The support splitting algorithm. Information Theory, IEEE Transactions on, 1193–1203 (2000) https://doi.org/10.1109/18.850662

[41] Sendrier, N., Symbolique, P.: On the dimension of the hull. SIAM Journal on Discrete Mathematics **10** (1995) https://doi.org/10.1137/S0895480195294027

[42] Budroni, A., Chi-Domínguez, J.-J., D'Alconzo, G., Di Scala, A.J., Kulkarni, M.: Don't use it twice! solving relaxed linear equivalence problems. In: Chung, K.-M., Sasaki, Y. (eds.) Advances in Cryptology – ASIACRYPT 2024, pp. 35–65. Springer, Singapore (2025)

[43] Leon, J.: Computing automorphism groups of error-correcting codes. IEEE Transactions on Information Theory **28**(3), 496–511 (1982) https://doi.org/10.1109/TIT.1982.1056498

[44] Beullens, W.: Not enough less: An improved algorithm for solving code equivalence problems over fq. In: Dunkelman, O., Jacobson, M.J. Jr., O'Flynn, C. (eds.) Selected Areas in Cryptography, pp. 387–403. Springer, Cham (2021)

[45] Biasse, J.-F., Micheli, G., Persichetti, E., Santini, P.: Less is more: Code-based signatures without syndromes. In: Nitaj, A., Youssef, A. (eds.) Progress in Cryptology - AFRICACRYPT 2020, pp. 45–65. Springer, Cham (2020)

[46] Schnorr, C., Euchner, M.: Lattice basis reduction: Improved practical algorithms and solving subset sum problems, vol. 66, pp. 68–85 (2006). https://doi.org/10.1007/3-540-54458-5_51

[47] Chen, Y., Nguyen, P.Q.: Bkz 2.0: Better lattice security estimates. In: Lee, D.H., Wang, X. (eds.) Advances in Cryptology – ASIACRYPT 2011, pp. 1–20. Springer, Berlin, Heidelberg (2011)

[48] Schnorr, C.: Lattice reduction by random sampling and birthday methods, vol. 2607 (2003). https://doi.org/10.1007/3-540-36494-3_14

[49] Alkim, E., Ducas, L., Pöppelmann, T., Schwabe, P.: Post-quantum key Exchange—A new hope. In: 25th USENIX Security Symposium (USENIX Security 16), pp. 327–343 (2016). https://www.usenix.org/conference/usenixsecurity16/technical-sessions/presentation/alkim

[50] Becker, A., Ducas, L., Gama, N., Laarhoven, T.: New directions in nearest neighbor searching with applications to lattice sieving. In: Proceedings of the Twenty-seventh Annual ACM-SIAM Symposium on Discrete Algorithms, pp. 10–24 (2016). SIAM

[51] Laarhoven, T., Mosca, M., Van De Pol, J.: Finding shortest lattice vectors faster using quantum search. Designs, Codes and Cryptography **77**, 375–400 (2015)

[52] Ackermann, L., Roux-Langlois, A., Wallet, A.: Public-Key Encryption from the Lattice Isomorphism Problem. In: WCC 2024 - The Thirteenth International Workshop on Coding and Cryptography, Perugia, Italy, pp. 1–11 (2024). https://inria.hal.science/hal-04924507